Université Sorbonne Université

Ecole Doctorale Informatique, Télécommunications et Electronique - ED130

Inria de Paris / Équipe ALMAnaCH

Thèse de doctorat

Discipline : Informatique

Présentée par

Rian TOUCHENT-SAAD

Pour obtenir le grade universitaire de

Docteur de l'Université Sorbonne Université

Improving Clinical Information Extraction with Public-Data Pretraining

---

Présentée et soutenue publiquement le DATE devant le jury composé de :

Mathieu Constant	ATILF (CNRS / Université de Lorraine)	Rapporteur
Aurélie Névéol	CNRS, LISN, Université Paris-Saclay	Rapportrice
Natalia Grabar	CNRS, Université de Lille	Rapportrice
Pierre Zweigenbaum	CNRS, LISN, Université Paris-Saclay	Examinateur
Thierry Charnois	LIPN, Université Sorbonne Paris Nord	Examinateur
Laurent Romary	Inria - ALMAnaCH	Directeur
Éric de la Clergerie	Inria - ALMAnaCH	Co-encadrant

Abstract

Contents

1 Introduction 1

I Related Works 3

2 Language Models 5
- 2.1 Introduction . . . 5
- 2.2 From Statistical to Neural Language Models . . . 5
  - 2.2.1 N-gram Models . . . 5
  - 2.2.2 Neural LMs . . . 5
  - 2.2.3 RNNs . . . 6
- 2.3 Transformer . . . 6
  - 2.3.1 Self-Attention . . . 6
  - 2.3.2 Positional Encodings . . . 6
  - 2.3.3 Encoder vs Decoder . . . 7
- 2.4 Pretraining Objectives . . . 7
  - 2.4.1 CLM (Causal Language Modeling) . . . 7
  - 2.4.2 MLM (Masked Language Modeling) . . . 7
  - 2.4.3 Objectifs hybrides (2024-2025) . . . 7
- 2.5 Scaling Laws & LLMs . . . 8
  - 2.5.1 Scaling . . . 8
  - 2.5.2 Post-ChatGPT . . . 8
- 2.6 Continual Pretraining . . . 9
  - 2.6.1 Fondements et terminologie . . . 9
  - 2.6.2 Catastrophic Forgetting . . . 9
  - 2.6.3 Solutions recentes (2024-2025) . . . 10
  - 2.6.4 From scratch vs continual PT . . . 10
  - 2.6.5 Encoders vs decoders: dynamiques differentes . . . 11
  - 2.6.6 Tentatives anterieures en francais biomedical . . . 11
- 2.7 Modeles pretrained biomed . . . 11
  - 2.7.1 Encoders biomed anglais . . . 11
  - 2.7.2 Modeles francais . . . 12
  - 2.7.3 Decoders biomed . . . 12
- 2.8 Architectures modernes . . . 13
  - 2.8.1 Modernisation des encoders . . . 13

2.8.2 Efficient Attention . . . 13
2.8.3 Long context . . . 13
2.9 Tokenization . . . 14
2.9.1 Methodes subword . . . 14
2.9.2 Domain mismatch . . . 14
2.10 Limites et transition . . . 15

3 Pretraining Data: Corpora and Knowledge 17
3.1 Introduction . . . 17
3.2 Web-Scale Pretraining Corpora . . . 17
3.2.1 Common Crawl . . . 17
3.2.2 Premiers grands corpus . . . 18
3.2.3 Corpus modernes (2024-2025) . . . 18
3.3 Data Quality and Curation . . . 19
3.3.1 Filtrage heuristique . . . 19
3.3.2 Classifieurs de qualite . . . 20
3.3.3 Organisation par domaine . . . 20
3.3.4 Deduplication . . . 21
3.3.5 Contamination . . . 21
3.4 Biomedical and Clinical Corpora . . . 21
3.4.1 Grandes sources anglophones . . . 21
3.4.2 Curation article-level pour le biomed . . . 22
3.4.3 Sources francophones . . . 23
3.5 Medical Ontologies and Knowledge Graphs . . . 24
3.5.1 Principales ontologies medicales . . . 24
3.5.2 Representations formelles . . . 25
3.6 Knowledge-Enhanced Pretraining . . . 25
3.6.1 Embeddings statiques de graphes . . . 25
3.6.2 Injection de connaissances dans les transformers . . . 26
3.6.3 Approches contrastives medicales . . . 27
3.6.4 Graph-to-text pour le pretraining . . . 27
3.7 Limites et transition . . . 28

4 Clinical Information Extraction 31
4.1 Introduction . . . 31
4.2 Named Entity Recognition . . . 32
4.2.1 Sequence Labeling . . . 32
4.2.2 BERT-based NER . . . 32
4.2.3 Au-dela du token-level . . . 33
4.3 Biomedical and Clinical NER . . . 33
4.3.1 Benchmarks anglais . . . 33
4.3.2 Benchmarks francais . . . 34

4.3.3 Specificites du texte clinique . . . 35
4.4 Medical Coding . . . 36
4.4.1 Le probleme . . . 36
4.4.2 Approches . . . 37
4.4.3 Codage en francais . . . 38
4.5 Entity Linking and Normalization . . . 38
4.5.1 Le probleme . . . 38
4.5.2 Approches . . . 39
4.5.3 Limites . . . 39
4.6 Zero-shot and Open NER . . . 40
4.6.1 Pourquoi zero-shot . . . 40
4.6.2 GLiNER . . . 40
4.6.3 Autres approches . . . 41
4.6.4 Limites du zero-shot actuel . . . 42
4.7 Synthetic Data for Clinical NLP . . . 42
4.7.1 Le probleme de la rarete . . . 42
4.7.2 LLM-as-annotator . . . 43
4.7.3 Generation de notes cliniques synthetiques . . . 43
4.8 Evaluation . . . 44
4.8.1 Metriques NER . . . 44
4.8.2 Impact de la tokenisation . . . 45
4.8.3 Besoin d'un benchmark unifie . . . 46
4.9 Limites et transition . . . 46

II Building a Biomedical Corpus 49

5 Collecting Biomedical Text 51
5.1 Introduction . . . 51
5.2 Existing French Biomedical Corpora . . . 52
5.3 The biomed-fr Corpus . . . 53
5.4 Vocabulary Analysis . . . 54
5.5 Limitations . . . 55

6 Detecting Content Types 57
6.1 Introduction . . . 57
6.2 Content Heterogeneity in Scientific Articles . . . 57
6.3 Method . . . 58
6.3.1 Data Collection and Preprocessing . . . 58
6.3.2 Two-Stage Annotation Framework . . . 59
6.3.3 Data Analysis . . . 60
6.3.4 Dataset Construction . . . 61

6.4 Experimental Setup . . . 61
6.4.1 Decoder Setup . . . 61
6.4.2 Encoder Setup . . . 62
6.4.3 Evaluation . . . 62
6.5 Results . . . 63
6.5.1 Decoder Results . . . 63
6.5.2 Encoder Results . . . 63
6.6 Discussion . . . 64
6.7 The Risks of Neural Quality Filtering . . . 66
6.8 Limitations . . . 68

7 Which Quality Signals Matter? 71
7.1 Introduction . . . 71
7.2 Sources . . . 72
7.3 Curation Pipeline . . . 73
7.4 Content-Type Ablation . . . 74
7.5 Quality Filtering and Upsampling . . . 74
7.6 Final Composition . . . 75

III Pretraining Language Models 77

8 Encoder Models for French Biomedicine 79
8.1 Introduction . . . 79
8.2 Pre-training . . . 79
8.3 Fine-tuning and Evaluation . . . 80
8.4 Results and Discussion . . . 80
8.5 Evaluating Models: Methodology Matters . . . 82
8.6 Environmental Impact . . . 84
8.7 Limitations . . . 84

9 Beyond Masked Language Modeling 85
9.1 Introduction . . . 85
9.2 Motivation . . . 85
9.2.1 When Decoder Continue-Pretraining Stops Working . . . 85
9.2.2 Architecture and Objective . . . 85
9.3 Method . . . 86
9.3.1 The CLM Detour . . . 86
9.3.2 Freeze Interventions . . . 87
9.3.3 Data . . . 87
9.3.4 Training Details . . . 88
9.3.5 Fine-tuning Protocol . . . 88

9.3.6 Evaluation . . . 88
9.4 Results . . . 89
9.4.1 French Biomedical Tasks . . . 89
9.4.2 English Biomedical Tasks . . . 89
9.4.3 Optimal Decay Ratio . . . 90
9.4.4 Needle-in-a-Haystack . . . 90
9.5 Analysis: Why Does the CLM Detour Work? . . . 92
9.5.1 CKA Methodology . . . 92
9.5.2 The CLM Imprint Persists in Low Layers . . . 92
9.5.3 Causal Evidence: Freeze Interventions . . . 93
9.5.4 Non-Localizability . . . 93
9.5.5 Scaling and Architectural Asymmetry . . . 94
9.6 Limitations . . . 95

10 Knowledge-Enriched Pretraining 97
10.1 Introduction . . . 97
10.2 Related Work . . . 98
10.2.1 Biomedical Pretrained Language Models . . . 98
10.2.2 Knowledge-Enhanced Language Models . . . 98
10.2.3 Automatic Medical Coding . . . 99
10.3 Method . . . 99
10.3.1 Medical Ontologies . . . 100
10.3.2 Ontology Walk Generation . . . 101
10.3.3 Textbook Generation . . . 101
10.3.4 OntoBook Training . . . 102
10.4 Experiments . . . 103
10.4.1 Experimental Setup . . . 103
10.4.2 Main Results . . . 104
10.4.3 Ablation Study . . . 105
10.4.4 Training Dynamics and Hyperparameter Sensitivity . . . 105
10.4.5 Multi-Ontology Transfer . . . 106
10.4.6 Probing Analysis . . . 107
10.5 Discussion . . . 107
10.6 Conclusion . . . 108

IV Adapting to Clinical Tasks 111

11 Limits of Direct Fine-Tuning 113

12 Architectures for Low-Resource Extraction 115

13 Synthetic Data for Task Adaptation 117

14 Conclusion 119

Acronyms 121

Glossary 123

1 Introduction

> "The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin. [...] We should build in only the meta-methods that can find and capture this arbitrary complexity. [...] The eventual success is tinged with bitterness, and often incompletely digested."
>
> - Richard S. Sutton, *The Bitter Lesson*

Healthcare has been one of the earliest and most prominent application domains for artificial intelligence. Use cases range from diagnostic support to medical record management, cohort selection, report generation, and clinical information extraction.

As early as the 1970s, MYCIN, the iconic expert system developed at Stanford, diagnosed bacterial infections and recommended antibiotics. It already achieved performance comparable to human specialists in its narrow domain. INTERNIST-1, later known as CADUCEUS, tackled internal medicine diagnosis across hundreds of diseases. However, these systems relied on explicit rules and static knowledge bases, which limited their ability to handle the complexity and variability of real-world medical data. They were rigid and required constant maintenance to keep up with medical advances. Building such systems demanded expensive medical experts to annotate clinical data and write rules by hand.

With the rise of machine learning in the 1990s and 2000s, data-driven approaches began to dominate healthcare AI. Supervised and unsupervised learning algorithms trained models directly from data. The interpretability of rule-based systems was traded for the performance of deep neural networks capable of capturing complex patterns. R2 ImageChecker (1998) became the first FDA-approved computer-aided detection system for mammography, using neural networks to spot suspicious micro-calcifications. PAPNET applied neural networks to cytological screening for cervical cancer.

Then came the scaling era. The recipe became simple: pretrain a language model on billions of words of raw text, then fine-tune on the target task. This approach routinely outperformed systems built with years of domain expertise. The need for expensive annotation, hand-crafted rules, and curated knowledge bases faded, beaten by data and compute.

Despite healthcare offering many structured and curated knowledge bases such as UMLS, SNOMED-CT, and RxNorm, experience has shown that general-purpose

language models pretrained on massive unstructured data outperform approaches based on domain-specific rules and knowledge bases. It seems our models cannot effectively integrate these structured resources, or perhaps these knowledge bases do not cover the full complexity and variability of medical knowledge as it is used in practice.

However, healthcare faces a limitation that other domains do not: the confidentiality and scarcity of annotated clinical data. Medical data is sensitive and difficult to obtain in large quantities for model training. Works adapting language models to the medical domain often rely on public data such as PubMed or patient forums, but these do not always reflect the real clinical language used in electronic health records. The clinical domain has unique jargon, abbreviations, and writing styles that differ from general medical texts. Moreover, while MIMIC exists in the United States, public clinical datasets remain scarce in other countries such as France.

This leads to a fundamental problem: the paradigm that has won is pretraining at scale, but in healthcare we lack massive public clinical data to pretrain language models. Can we find new ways to exploit public data for pretraining language models adapted to the clinical domain, without using sensitive clinical data?

Part I

Related Works

2 Language Models

2.1 Introduction

Un language model c'est un modele probabiliste sur des sequences de tokens:

$$P(\mathbf{w}) = \prod_{t=1}^{L} P(w_t \mid w_{<t})$$

Factorisation autoregressive (chain rule). Fondation de tout le NLP moderne. Pourquoi c'est important pour la these:

- Clinical IE repose sur des LMs pretrained
- Domain adaptation = defi central
- French + biomedical + clinical = triple rarete de donnees

-> plan du chapitre: on remonte des n-grams jusqu'aux archis modernes, puis on zoome sur le continual pretraining qui est au coeur de la these

2.2 From Statistical to Neural Language Models

2.2.1 N-gram Models

Hypothese de Markov: $P(w_t \mid w_{<t}) \approx P(w_t \mid w_{t-n+1}, \dots, w_{t-1})$.

- Kneser-Ney smoothing
- Probleme: curse of dimensionality, $|\mathcal{V}|^n$ explose

2.2.2 Neural LMs

- Bengio et al. (2003): neural probabilistic LM
  - Embeddings denses, resout la sparsity
  - Mots similaires -> vecteurs proches
- Word2Vec (Mikolov et al., 2013): skip-gram / CBOW
  - $\vec{\text{king}} - \vec{\text{man}} + \vec{\text{woman}} \approx \vec{\text{queen}}$

- ELMo (Peters et al., 2018): embeddings contextuels via biLSTM
  - Meme mot, contextes differents -> vecteurs differents
  - Precurseur du pretrain-then-finetune

2.2.3 RNNs

- LSTM (Hochreiter & Schmidhuber, 1997)
- En theorie: long-range dependencies
- En pratique: vanishing gradients, pas parallelisable

-> bottleneck pour scaler -> besoin d'une nouvelle archi

2.3 Transformer

2.3.1 Self-Attention

Vaswani et al. (2017).

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

- Interactions directes cross-position (vs RNN indirect via hidden state)
- $O(L^2)$ temps et memoire
- Parallelisable
- Multi-head: h tetes, chacune specialise (syntaxe, semantique, position)

2.3.2 Positional Encodings

- APE sinusoidal (Vaswani 2017) -> extrapolation limitee
- APE learned (Devlin 2019) -> pas d'extrapolation
- ALiBi (Press et al. 2022) -> bias lineaire, mieux
- RoPE (Su et al. 2021) -> matrices de rotation, le meilleur

RoPE = standard maintenant: LLaMA, Mistral, ModernBERT.

2.3.3 Encoder vs Decoder

- Encoder: attention bidirectionnelle, MLM, classification/NER (BERT, RoBERTa)
- Decoder: attention causale, CLM, generation (GPT, LLaMA)
- Encoder-decoder: T5 (Raffel et al., 2020), seq2seq, plus trop utilise

-> maintenant qu'on a l'archi, question: quel objectif de pretraining?

2.4 Pretraining Objectives

2.4.1 CLM (Causal Language Modeling)

GPT (Radford et al., 2018).

$$\mathcal{L}_{\text{CLM}} = -\sum_{t=1}^{L} \log P(w_t \mid w_{<t})$$

- Tous les tokens contribuent au gradient
- Supervision dense
- Naturel pour la generation

2.4.2 MLM (Masked Language Modeling)

BERT (Devlin et al., 2019).

$$\mathcal{L}_{\text{MLM}} = -\sum_{t \in \mathcal{M}} \log P(w_t \mid \mathbf{w}_{\setminus t})$$

- 15% tokens masked (80% [MASK], 10% random, 10% unchanged)
- Contexte bidirectionnel -> representations plus riches
- Mais: seulement 15% contribuent -> gradient sparse
- Contraste avec CLM ou chaque token compte

2.4.3 Objectifs hybrides (2024-2025)

Important pour le Ch.5 de la these.
Approche CLM puis MLM:

- CLM d'abord puis decay vers MLM = meilleur que MLM seul
- CLM: gradients denses (tous les tokens), MLM: sparse (15%)

- Hypothese: CLM "compresse" la connaissance, MLM restaure la bidirectionnalite

Refs:

- "Should We Still Pretrain Encoders with MLM?" (arXiv:2507.00994, 2025) -> biphasic CLM vers MLM bat pure MLM
- GPT-BERT (arXiv:2410.24159, 2024) -> unifie CLM+MLM
- AntLM (CoNLL 2024) -> alternance CLM/MLM

Note: ELECTRA (Clark et al., 2020) avait propose RTD pour signal plus dense, mais ModernBERT montre que bonne archi + MLM suffit (bat DeBERTaV3 qui utilise RTD).
-> on a les archis et les objectifs, question maintenant: est-ce que ca scale?

2.5 Scaling Laws & LLMs

2.5.1 Scaling

- GPT-3 (Brown et al., 2020): 175B params, few-shot emergent, prompting remplace le finetuning
- Chinchilla (Hoffmann et al., 2022):

$$L(N, D) = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + E$$

  -> compute-optimal: smaller model + more data bat larger model + less data
- LLaMA (Touvron et al., 2023): 7B rivalise avec bcp plus gros via 1T tokens -> open weights, democratisation

2.5.2 Post-ChatGPT

- RLHF (Ouyang et al., 2022) -> preference learning
- Instruction tuning
- ChatGPT, Claude, Gemini
- Emergent abilities (Wei et al., 2022) -> chain-of-thought etc.

-> ok ca scale pour le general, mais nous on a un domaine specifique -> comment adapter?

2.6 Continual Pretraining

Section centrale pour la these. C'est le coeur de ce qu'on fait (CamemBERT-bio, ModernCamemBERT-bio).

2.6.1 Fondements et terminologie

"Don't Stop Pretraining" (Gururangan et al., ACL 2020):

- DAPT = Domain-Adaptive Pre-Training: 2eme phase sur corpus domaine
- TAPT = Task-Adaptive Pre-Training: sur donnees non-annotees de la tache
- DAPT + TAPT combine -> meilleurs resultats
- Jusqu'a +3 F1 meme avec petit corpus

Taxonomie moderne (Wang et al., CSUR 2025):

- CPT = nouvelles donnees generales au fil du temps
- DAP = specialisation domaine
- CFT = adaptation sequentielle de taches

2.6.2 Catastrophic Forgetting

Le probleme (Kirkpatrick et al., 2017):

- Les reseaux ecrasent les poids necessaires aux taches precedentes
- EWC (Elastic Weight Consolidation): penaliser les changements sur les poids importants

Pourquoi ca compte pour les LLMs:

- Continual PT peut degrader les capacites generales
- Capacite a suivre des instructions particulierement fragile
- Petits modeles plus sensibles (Yildiz et al., 2024)

"Spurious forgetting" (2024): les baisses de perf refletent peut-etre une perte d'alignement, pas une vraie perte de connaissance.
-> ok c'est un vrai probleme, quelles solutions?

2.6.3 Solutions recentes (2024-2025)

Ibrahim et al. (2024), "Simple and Scalable Strategies":

- LR re-warming: relancer le learning rate quand on ajoute des donnees
- LR re-decaying: cosine decay apres warmup
- Replay: mixer 1% des donnees precedentes -> ca suffit
- Valide a 405M et 10B
- -> egalise from-scratch pour une fraction du compute

Stability gap (arXiv:2406.14833, 2024):

- Chute temporaire au debut du continual PT, puis recovery
- Solutions: subset propre, sous-corpus de qualite, data mixing
- 36.2% -> 40.7% avec seulement 40% du budget training

Data selection:

- Donnees non pertinentes = degradation
- 10% du corpus bien selectionne fait aussi bien que 100% vanilla continual PT

2.6.4 From scratch vs continual PT

- From scratch (PubMedBERT, DrBERT): vocab custom, pas de forgetting, mais tres cher
- Continual (BioBERT, Gururangan): efficient, preserve le general, mais tokenizer sous-optimal

Cout compute:

- DrBERT from scratch: 128x V100, 20h
- AliBERT from scratch: 48x A100, 20h
- BioBERT continual: 8x V100, ~10 jours
- -> from-scratch coute ~30x plus de compute

2.6.5 Encoders vs decoders: dynamiques differentes

Decoders (LLMs), ca marche pas bien:

- BioMistral: -0.9 points apres continual PT
- Dorfner et al. (2024): biomedical LLMs sous-performent les generalistes
- OpenBioLLM-8B: 30% vs Llama-3-8B: 64.3% sur cas NEJM
- -> hypothese: perte de connaissances generales pendant la specialisation

Encoders, c'est different:

- Contexte bidirectionnel peut-etre plus robuste au forgetting
- MLM moins sujet au catastrophic forgetting?
- -> voir Ch.5: approche CLM vers MLM sur les encoders

-> tension narrative: continual PT marche pas pour les decoders, et pour les encoders? -> c'est la question de la these

2.6.6 Tentatives anterieures en francais biomedical

- Copara et al. (2020): 31K articles -> aucune amelioration (corpus trop petit)
- Le Clercq de Lannoy et al. (2022): 136M mots -> +2 EMEA seulement
- Dura et al. (2022): 21M docs APHP -> +3% mais donnees privees

Question ouverte: pourquoi des resultats contradictoires? DrBERT dit que continual PT marche pas. CamemBERT-bio (cette these) montre que si, avec le bon corpus.

-> ok on sait ce qu'est le continual PT, maintenant quels modeles existent?

2.7 Modeles pretrained biomed

2.7.1 Encoders biomed anglais

- BioBERT (Lee et al., 2019): PubMed+PMC 18B mots, continual -> +0.62% NER, +2.80% RE
- SciBERT (Beltagy et al., 2019): Semantic Scholar 1.14M papers, from scratch -> 42% vocab overlap avec BERT
- PubMedBERT (Gu et al., 2022): PubMed only, from scratch -> introduit BLURB benchmark

- ClinicalBERT (Alsentzer et al., 2019): MIMIC-III, continual -> acces restreint
- BioLinkBERT (Yasunaga et al., 2022): PubMed + citation links, from scratch -> +7% BioASQ

SOTA 2025: BioClinical ModernBERT, cf section architectures modernes.

2.7.2 Modeles francais

General:

- CamemBERT (Martin et al., 2020): RoBERTa sur OSCAR 138GB -> resultat interessant: 4GB donne a peu pres la meme perf que 138GB
- FlauBERT (Le et al., 2020)
- CamemBERT 2.0 / CamemBERTa (2024): architecture DeBERTaV3, nouveau tokenizer

Biomedical francais:

- DrBERT (Labrak et al., 2023): from scratch, public + prive -> pretend que continual PT marche pas pour le francais biomed
- AliBERT (Berhe et al., 2023): from scratch, Unigram regularise -> pas disponible publiquement

-> tension: DrBERT dit from-scratch necessaire, CamemBERT-bio (cette these) montre le contraire

2.7.3 Decoders biomed

- BioMistral (Labrak et al., 2024): 3B tokens PMC, 32x A100 20h, base Mistral
- Meditron (Chen et al., 2023): 46B tokens, 128x A100 332h, base Llama-2
- PMC-LLaMA (Wu et al., 2024): 75B tokens, massif

Continual PT pour decoders = resultats mitiges:

- BioMistral: -0.9 points, recupere seulement via model merging
- Dorfner et al. (2024): "Biomedical LLMs Seem not to be Superior to Generalist Models"
- -> contraste avec les encoders, voir Ch.5

-> les modeles existent mais les archis ont evolue, quoi de neuf?

2.8 Architectures modernes

2.8.1 Modernisation des encoders

DeBERTa (He et al., ICLR 2021):

- Disentangled attention: separe contenu et position
- Premier a battre l'humain sur SuperGLUE
- -> influence sur ModernBERT, CamemBERT 2.0

MosaicBERT (Portes et al., NeurIPS 2023):

- 30% masking au lieu de 15% BERT
- FlashAttention + ALiBi + GLU + unpadding
- BERT-base en 1.13h sur 8x A100 (~$20)
- -> fondation directe de ModernBERT

2.8.2 Efficient Attention

FlashAttention (Dao et al., 2022):

- Reorganise le calcul pour la hierarchie memoire GPU (tiling, kernel fusion)
- Meme $O(L^2)$ mais 2-4x plus rapide en pratique
- Standard maintenant

Autres:

- GQA (Ainslie et al., 2023): partage de KV-heads -> Llama-2, Mistral
- RMSNorm (Zhang & Sennrich, 2019): LayerNorm simplifie

2.8.3 Long context

Le probleme: BERT = 512 tokens, comptes-rendus cliniques = souvent 2000+. Tronquer = perdre de l'info diagnostique.

- Longformer (Beltagy et al., 2020): 4096 tokens, sliding window + global attention
- BigBird (Zaheer et al., 2020): 4096, sparse attention
- Clinical Longformer (Li et al., 2022): 4096, Longformer continue pretrained sur MIMIC-III

- ModernBERT (Warner et al., 2024): 8192 tokens
  - FlashAttention v2/v3
  - RoPE
  - Alternance local/global attention
  - 30% masking (de MosaicBERT)
  - -> 16x le contexte de BERT

Adaptations cliniques 2025:

- Clinical ModernBERT (Lee et al., 2025): PubMed + MIMIC-IV + ontologies
- BioClinical ModernBERT (2025): 53.5B tokens, 20 datasets cliniques -> SOTA actuel

2.9 Tokenization

2.9.1 Methodes subword

- BPE (Sennrich et al., 2016): merge les paires les plus frequentes
- SentencePiece (Kudo & Richardson, 2018): language-independent
- Unigram (Kudo, 2018): selection probabiliste

2.9.2 Domain mismatch

SciBERT: vocab general vs scientifique = 42% intersection seulement.
Exemple biomed francais:

- Tokenizer general: “echocardiographie” -> [“echo”, “##cardi”, “##ographie”]
- Tokenizer domaine: “echocardiographie” -> [“echocardiographi”, “e”]

Tradeoff:

- From-scratch = vocab custom mais cher
- Continual PT = tokenizer sous-optimal mais efficient

2.10 Limites et transition

- Rarete des donnees: texte clinique prive (CNIL), modeles hospitaliers pas partageables -> besoin d'alternatives publiques -> Ch.3 Corpus Annotation
- Knowledge gap: LMs captent des patterns, pas la structure des ontologies. Codage medical = connaissances hierarchiques -> approches knowledge-enhanced -> Ch.4 Clinical IE
- Documents longs: comptes-rendus hospitaliers depassent les limites de BERT. Longformer et ModernBERT aident mais faut adapter au domaine

-> ce chapitre a couvert les fondations techniques du language modeling. Le chapitre suivant traite de comment construire et annoter des corpus pour entrainer ces modeles.

3 Pretraining Data: Corpora and Knowledge

3.1 Introduction

Le chapitre precedent couvrait les modeles. Maintenant faut les nourrir.

Les LMs modernes sont data-hungry:

- GPT-3 (Brown et al., 2020): 300B tokens
- LLaMA (Touvron et al., 2023): 1-2T tokens
- Chinchilla (Hoffmann et al., 2022): compute-optimal = plus de data, modele plus petit

La qualite et la composition du corpus de pretraining determinent largement les capacites du modele. "Data is the new bottleneck."

Deux types de donnees de pretraining:

- Texte brut: web crawls, articles scientifiques, livres
- Connaissances structurees: ontologies medicales, knowledge graphs

Pourquoi c'est central pour la these:

- Part 1 = construire un corpus biomed francais a partir de sources publiques
- Ch.6 = injecter des connaissances d'ontologies medicales pendant le pretraining
- Triple rarete: francais + biomedical + clinique

-> plan du chapitre: on part des corpus web generalistes, on zoome sur le filtrage qualite, puis sur les sources biomed specifiques, et enfin sur les ontologies medicales et comment les integrer dans le pretraining

3.2 Web-Scale Pretraining Corpora

3.2.1 Common Crawl

Source de base de presque tous les grands corpus de pretraining:

- Web crawl ouvert, 2.5B pages par snapshot mensuel, 250-400 TiB non compresse
- Snapshots mensuels depuis 2008, petaoctets cumules
- Contenu tres heterogene: articles, forums, spam, code, pubs, boilerplate...
- Pas utilisable directement -> necessite des pipelines de nettoyage lourdes

3.2.2 Premiers grands corpus

C4 (Raffel et al., 2020):

- Colossal Clean Crawled Corpus, 750GB d'anglais extrait d'un seul snapshot CC (avril 2019)
- Filtres: langdetect >= 99% anglais, suppression phrases incompletes, dedup
- Dodge et al. (2021) documentent le contenu: brevets, sites militaires US, texte machine-generated, exemples de benchmarks
- Blocklist filtering retire disproportionnellement du texte de/sur les minorites

The Pile (Gao et al., 2020):

- 800GB anglais, 22 sources diversifiees (PubMed, ArXiv, GitHub, StackExchange, Wikipedia, livres, etc.)
- Idee: diversite des sources > volume brut d'une seule source
- Open-source (EleutherAI)

OSCAR (Ortiz Suarez et al., 2019):

- Pipeline asynchrone pour classifier CC par langue
- Optimise pour infra moyenne/basse ressource (I/O-bound)
- Produit le corpus de CamemBERT: 138GB de texte francais
- Shuffled au niveau ligne pour eviter les problemes de copyright (ironiquement)

3.2.3 Corpus modernes (2024-2025)

FineWeb (Penedo et al., 2024):

- 15T tokens, 96 snapshots CC
- Documentation detaillee des strategies de dedup et filtrage

- Pipeline: extraction texte, filtrage langue, heuristiques qualite, dedup URL + MinHash
- FineWeb-Edu: sous-ensemble 1.3T tokens filtre pour contenu educatif (cf section suivante)

RedPajama-V2 (Weber et al., 2024):

- Dataset ouvert pour entrainer des LLMs
- Inclut subset francais (RPv2-Fr)
- Metriques de qualite pre-calculees pour faciliter le filtrage

TxT360 (Tang et al., 2024):

- "Perfect blend" de sources
- Dedup globale sur 99 snapshots CC: 20TB -> 4.83T tokens (reduction 80%)
- Pipeline similaire a FineWeb + near-dedup globale additionnelle

Autres:

- Dolma (Soldaini et al., 2024): 3T tokens, corpus ouvert pour recherche (OLMo)
- ROOTS (Laurencon et al., 2023): 1.6TB, multilingue composite (BigScience/BLOOM)

-> on a des milliards de tokens de texte web, mais la qualite varie enormement -> comment selectionner les bons documents?

3.3 Data Quality and Curation

3.3.1 Filtrage heuristique

Approches classiques, devenues standard depuis Gopher (Rae et al., 2022):

- Regles: longueur min/max du document, ratio mots/symboles, proportion de majuscules, lignes dupliquees
- Filtrage par perplexite: KenLM entraine sur Wikipedia, documents a perplexite trop haute = bruit
- Language ID: fastText pour filtrer par langue
- Ces heuristiques sont reprises par presque tous les pipelines modernes (FineWeb, C4, Dolma)

Limites: les heuristiques sont necessaires mais insuffisantes. Elles retirent le bruit evident mais ne distinguent pas la qualite du contenu.

3.3.2 Classifieurs de qualite

Paradigme recent: LLM-as-annotator -> distillation vers petit classifieur -> filtrage a grande echelle.

FineWeb-Edu (Penedo et al., 2024):

- Llama-3-70B-Instruct annote 460K pages web sur "educational value" (echelle 0-5)
- Echelle additive: le LLM evalue chaque critere et construit le score pas a pas
- Prompt cible le niveau primaire/college pour eviter de favoriser les papiers trop techniques
- Distillation vers classifieur de regression (Snowflake-arctic-embed, F1 = 82% en binaire avec seuil >= 3)
- Classification de 15T tokens: 6K heures H100
- Retire 92% des donnees, bat quand meme le corpus complet sur MMLU, ARC, OpenBookQA

DCLM / DataComp-LM (Li et al., 2024):

- Classifieur FastText entraine pour separer instructions synthetiques (Open Hermes 2.5) du web general
- Favorise les structures Q&A resolues: question courte + reponse directe + raisonnement bref
- Utilise dans le mix de pretraining d'OLMo2

Nemotron-CC (NVIDIA, 2024):

- Classifieur de qualite multi-criteres (accuracy, clarity, coherence, etc.)
- Approche plus granulaire que le score educatif unique de FineWeb-Edu

3.3.3 Organisation par domaine

WebOrganizer (Wettig et al., 2025):

- Organise le web en taxonomies par topic et par format
- Annotations LLM distillees vers classifieurs efficaces
- Montre que FineWeb-Edu a un biais implicite vers Science & Technology et Academic Writing
- L'efficacite de FineWeb-Edu vient en partie de preferences de domaine alignees avec les benchmarks (MMLU, HellaSwag)
- Combiner organisation par domaine + filtrage qualite > qualite seule

3.3.4 Deduplication

- URL-level: retirer les duplicats exacts d'URL
- Document-level: MinHash + LSH pour near-duplicates
- TxT360: dedup globale sur 99 snapshots CC, reduction de 80%
- Crucial: la duplication biaise le training vers les textes les plus copies (boilerplate, templates)

3.3.5 Contamination

Le filtrage agressif pose un probleme de contamination des benchmarks:

- Deng et al. (2024): ChatGPT et GPT-4 devinent 52% et 57% des options manquantes de MMLU -> les QA pairs sont sur le web
- Xu et al. (2024): survey des techniques de detection de contamination dans les LLMs
- InfiniGram (Liu et al., 2024): outil pour identifier des exact matches dans les training sets

Tension fondamentale: filtrer plus agressivement = meilleurs scores benchmarks, mais potentiellement via contamination implicite plutot que vraie qualite.

-> ok le filtrage marche pour les corpus web generalistes, mais en biomedical c'est different: les corpus sont plus petits, le domaine est tres specifique, les criteres de qualite ne sont pas les memes -> quelles sources biomed existent?

3.4 Biomedical and Clinical Corpora

3.4.1 Grandes sources anglophones

PubMed:

- Base de donnees de la NLM (National Library of Medicine)
- 39M citations et abstracts d'articles biomedicaux
- Texte structure, vocabulaire technique, qualite editoriale
- Abstracts seulement (pas le full-text)
- Utilise par BioBERT (Lee et al., 2019), PubMedBERT (Gu et al., 2022)

PMC Open Access:

- 4.5M articles full-text en acces libre
- 98% anglais (Labrak et al., 2024)
- Contenu heterogene: articles de recherche, reviews, case reports, editoriaux, guidelines
- Utilise par BioMistral (3B tokens), Meditron (46B tokens), PMC-LLaMA (75B tokens)
- Probleme: granularite article. Un article de recherche peut contenir a la fois des cas cliniques pertinents et des sections methodologiques non pertinentes

MIMIC (Johnson et al., 2016/2023):

- MIMIC-III, MIMIC-IV: notes cliniques du Beth Israel Deaconess Medical Center
- Acces restreint (data use agreement, formation ethique)
- Utilise par ClinicalBERT (Alsentzer et al., 2019), Clinical Longformer (Li et al., 2022), BioClinical ModernBERT (Sounack et al., 2025)
- Gold standard pour le texte clinique anglais, mais un seul hopital americain -> pas generalizable

3.4.2 Curation article-level pour le biomed

Meditron (Chen et al., 2023):

- Score 0-1 par article base sur: MeSH tags, type de publication, reputation du journal, recence, nombre de citations
- Upsampling des articles a haut score, filtrage des bas scores
- 46B tokens au total, 128x A100, 332h pour le 70B
- Limite: granularite article. Un article peut melanger contenu pertinent et non-pertinent

BioClinical ModernBERT (Sounack et al., 2025):

- 169B tokens: PubMed + PMC + 20 datasets cliniques (MIMIC-III, MIMIC-IV, CheXpert Plus, etc.)
- SOTA actuel sur les benchmarks biomed anglais
- Mais: la plupart des sources cliniques necessitent des data use agreements -> pas vraiment "public"

3.4.3 Sources francophones

Ressources publiques existantes:

- ISTEX: base de 27M references scientifiques, permet d'extraire des documents francais bio/med
- CLEAR (Grabar & Cardon, 2018): notices medicamenteuses, articles d'encyclopedie medicale en francais technique et simplifie
- E3C (Magnini et al., 2021): corpus multilingue de cas cliniques, notices, resumes de these medicale
- HAL / Halvest (Kulumba et al., 2024): articles scientifiques francais et anglais extraits du depot HAL (Hyper Articles en Ligne)
- Wikipedia biomedical: articles des portails medecine, pharmacie, biologie. Libre et multilingue

Modeles entraines from scratch sur corpus francais:

- DrBERT (Labrak et al., 2023): corpus de 7GB, from scratch sur 128x V100
  - Pretend que le continual PT ne marche pas pour le francais biomed
  - Mais: evaluation potentiellement biaisee (cf Ch.4 de la these)
- AliBERT (Berhe et al., 2023): ScienceDirect + theses Sudoc, tokenizer Unigram regularise, 48x A100
  - Modele non disponible publiquement

Tentatives de continual pretraining en francais biomed:

- Copara et al. (2020): 31K articles seulement -> aucune amelioration sur CamemBERT-large
- Le Clercq de Lannoy et al. (2022): PubMed + Cochrane + ISTEX + Wikipedia = 136M mots -> +2 F1 sur EMEA, rien sur MEDLINE
- Dura et al. (2022): 21M documents cliniques APHP -> +3% sur APMed (prive), mais donnees non partageables

Resultats décevants: DrBERT dit que continual PT marche pas, Copara et Le Clercq confirment. Mais le probleme est peut-etre le corpus (trop petit, pas assez cible), pas la methode.

Triple rarete: francais + biomedical + clinique. Les textes cliniques sont prives (CNIL), les modeles hospitaliers ne sont pas partageables entre etablissements. Les sources publiques francophones sont fragmentees et petites comparees a PubMed/PMC.

-> le texte biomedical aide pour le pretraining, mais les LMs captent des patterns statistiques, pas la structure des ontologies. Pour le codage medical (CIM-10, CCAM) il faut des connaissances hierarchiques explicites -> peut-on utiliser directement les ontologies comme donnees de pretraining?

3.5 Medical Ontologies and Knowledge Graphs

3.5.1 Principales ontologies medicales

UMLS (Lindberg et al., 1993):

- Unified Medical Language System, meta-ontologie biomedical de la NLM
- Integre 189 vocabulaires sources (SNOMED-CT, MeSH, ICD, ATC, etc.)
- 3.4M concepts, 16.7M noms de concepts uniques
- Standard international pour la recherche en NLP biomed
- Contient: synonymes, relations hierarchiques, relations associatives, definitions

SNOMED-CT:

- 371K concepts cliniques structures
- Relations hierarchiques (is-a) + associatives (finding site, causative agent, etc.)
- Standard pour les systemes d'information clinique internationaux
- Couverture: diagnostics, procedures, anatomie, substances, organismes

CIM-10 / ICD-10 (OMS, 2019):

- Classification internationale des maladies
- Hierarchie: chapitres -> blocs -> categories -> sous-categories
- Version francaise enrichie par l'ATIH avec: exclusions, inclusions, notes, libelles courts/longs
- Utilise pour le codage PMSI (Programme de Medicalisation des Systemes d'Information) en France
- 19K codes dans la version ATIH

CCAM (ATIH):

- Classification Commune des Actes Medicaux, 8K codes

- Specifique a la France, structure hierarchique
- Relations: actes associes, incompatibilites, modificateurs

ATC (OMS):

- Anatomical Therapeutic Chemical, 6K codes au 5eme niveau
- 5 niveaux: anatomique -> therapeutique -> pharmacologique -> chimique -> substance
- Standard pour la classification des medicaments

3.5.2 Representations formelles

- RDF (Resource Description Framework): triplets sujet-predicat-objet, standard W3C
- OWL (Web Ontology Language): logique descriptive, raisonnement automatique, classes, proprietes
- SKOS (Simple Knowledge Organization System): prefLabel, altLabel (synonymes), broader/narrower (hierarchie), related, exactMatch

Les ontologies codent explicitement ce que les LMs doivent apprendre implicitement:

- Hierarchie: E11 (diabete type 2) $\subset$ E10-E14 (diabete sucre)
- Exclusions: E11 $\neq$ E10 (diabete type 1) – a ne pas confondre
- Synonymes: un meme concept a plusieurs libelles (prefLabel + altLabel)
- Relations associatives: complications, traitements, liens entre ontologies

-> les ontologies sont riches en connaissances structurees, mais sous forme de graphes: pas directement exploitables par les LMs qui attendent du texte en language natu -> comment les integrer dans le pretraining?

3.6 Knowledge-Enhanced Pretraining

3.6.1 Embeddings statiques de graphes

Approches qui produisent des vecteurs fixes (pas contextuels) a partir de graphes:

- Node2Vec (Grover & Leskovec, 2016): random walks sur graphe -> skip-gram, apprentissage de representations de noeuds

- RDF2Vec (Ristoski & Paulheim, 2016): adapte le paradigme walk + skip-gram aux graphes RDF. Fondation des approches de walks sur ontologies

- Snomed2Vec (Agarwal et al., 2019): Node2Vec applique a SNOMED-CT, 5-6x amelioration sur concept similarity

- OWL2Vec* (Chen et al., 2021): walks sur ontologies OWL (teste sur Gene Ontology)

Limite: embeddings statiques, pas integres dans un transformer. Un concept = un vecteur fixe quel que soit le contexte. Pas de contextualisation.

-> on peut faire mieux en integrant les KG directement dans les transformers

3.6.2 Injection de connaissances dans les transformers

Premiere generation: injecter des entites dans l'architecture:

- ERNIE (Zhang et al., 2019): aligne entity embeddings (TransE) avec representations textuelles pendant le pretraining. Entity linking implicite

- KnowBERT (Peters et al., 2019): integre des entity linkers comme couches d'attention supplementaires dans BERT

- K-BERT (Liu et al., 2020): injecte des triples de KG comme soft-position embeddings dans l'input. Modifie la sequence d'entree directement

Deuxieme generation: multi-task pretraining language + KG:

- KEPLER (Wang et al., 2021): MLM + TransE knowledge embedding sur Wikidata simultanement. Les deux objectifs partagent le meme encoder

- CoLAKE (Sun et al., COLING 2020): construit des word-knowledge graphs qui unifient contexte textuel et contexte KG, puis pretraine avec MLM modifie sur ces graphes

- DRAGON (Yasunaga et al., NeurIPS 2022): joint pretraining bidirectionnel MLM + link prediction sur UMLS
  - PubMed pour le texte, UMLS pour le KG
  - +3% accuracy sur MedQA, +10% sur raisonnement multi-step
  - Reference pour le multi-task MLM + KG

3.6.3 Approches contrastives medicales

- SapBERT (Liu et al., NAACL 2021): self-alignment pretraining sur synonymes UMLS
  - Contrastive learning: synonymes UMLS = positifs, non-synonymes = negatifs
  - Metric learning scalable sur 4M+ concepts UMLS
  - SOTA entity linking biomed sur 6 benchmarks
  - Un seul modele pour tous les types d'entites ("one-model-for-all")
- CODER (Yuan et al., 2022): contrastive pretraining sur 87M triplets de relations UMLS
  - Relations + synonymes dans l'objectif contrastif
  - Medical term normalization

3.6.4 Graph-to-text pour le pretraining

Approche recente: transformer les graphes en texte lisible par les LMs.
"Textbooks Are All You Need" (Gunasekar et al., 2023):

- Donnees synthetiques de qualite textbook generees par GPT -> petits modeles tres performants
- Montre que la qualite et la structure des donnees importent plus que le volume brut
- Paradigme: generer des donnees synthetiques structurees plutot que crawler du texte brut

AntGLM-Med (Zhou et al., 2023):

- Reecrit des sous-graphes de KG medical en prose naturelle
- Utilise pour le continual pretraining de LLMs medicaux
- Plus proche precedent de la generation de texte a partir d'ontologies pour pretraining

EntiGraph (Yang et al., ICLR 2025):

- Algorithme d'augmentation entity-centric: decompose un corpus en entites puis genere du texte sur les relations entre entites
- 1.3M tokens reels -> 600M tokens synthetiques via GPT-4-turbo
- Scaling log-lineaire de l'accuracy avec le nombre de tokens synthetiques

- Cible les decoders (Llama 3 8B), pas encore teste sur encoders

Limites actuelles du knowledge-enhanced pretraining:

- La plupart des travaux sont en anglais et sur UMLS
- Graph-to-text pour le pretraining est recent (2023-2025)
- Pas encore explore pour les encoders bidirectionnels (DRAGON fait du link prediction, pas du graph-to-text)
- Les ontologies nationales (CIM-10 ATIH, CCAM) n'ont jamais ete utilisees pour du pretraining
- SapBERT et CODER exploitent UMLS mais uniquement via contrastive learning, pas via generation de texte

-> methodes prometteuses mais gap important: langues non-anglaises, ontologies nationales, pretraining d'encoders via graph-to-text

3.7 Limites et transition

- Rarete biomed non-anglais: PMC = 98% anglais, les corpus francais biomed sont petits et fragmentes. Les pipelines de filtrage (FineWeb-Edu, DCLM) sont concus pour l'anglais -> besoin de strategies de collection et de filtrage adaptees au domaine et a la langue
- Granularite de filtrage: les approches existantes (Meditron, FineWeb-Edu) filtrent au niveau document ou article. Mais un article biomed peut contenir a la fois des cas cliniques pertinents et des sections non pertinentes -> besoin de filtrage a granularite plus fine (paragraphe)
- Criteres de qualite non adaptes: les criteres "educatifs" de FineWeb-Edu ne correspondent pas au texte biomedical (cas cliniques, notices medicamenteuses, protocoles). Les biais vers Science & Technology identifies par WebOrganizer peuvent etre differents en biomed -> besoin de criteres domaine-specifiques
- Ontologies nationales sous-exploitees: UMLS domine la recherche, mais les systemes hospitaliers francais utilisent CIM-10/ATIH, CCAM, ATC. Aucun travail n'a integre ces ontologies dans le pretraining -> gap pour le codage medical en francais
- Graph-to-text naissant: AntGLM-Med et EntiGraph montrent la voie mais sont limites aux decoders en anglais. Pas encore explore pour les encoders bidirectionnels ni pour les ontologies non-anglophones -> possibilite de generer du texte "textbook" a partir des ontologies ATIH

-> ce chapitre a couvert les donnees de pretraining: sources textuelles web et biomed, methodes de curation et filtrage, ontologies medicales et leurs integrations dans le pretraining. Le chapitre suivant traite de la tache finale: l'extraction d'information clinique, et de comment les modeles pretrained sont utilises pour cette tache.

4 Clinical Information Extraction

4.1 Introduction

Clinical Information Extraction (IE) = extraire de l'information structuree a partir de texte clinique non structure.

Pourquoi c'est important:

- Les entrepots de donnees cliniques (CDW) se developpent dans les hopitaux
- Raghavan et al. (2014): jusqu'a 80% des entites cliniques sont absentes des donnees structurees -> le texte libre (comptes-rendus, courriers, prescriptions) est la source principale
- En France: PMSI (Programme de Medicalisation des Systemes d'Information), chaque sejour hospitalier doit etre code avec des codes CIM-10 et CCAM. 30M sejours/an (ATIH)
- Le codage est fait manuellement par des codeurs professionnels -> automatisation = enjeu majeur

Types de taches:

- Named Entity Recognition (NER): identifier les mentions d'entites dans le texte
- Relation Extraction (RE): extraire les relations entre entites
- Medical coding: assigner des codes d'ontologie (CIM-10, CCAM, ATC) a un document
- Entity linking / normalization: associer une mention a un concept dans une base de connaissances (UMLS, SNOMED-CT)

Pourquoi c'est central pour la these:

- Les taches de clinical IE sont le terrain d'evaluation final des modeles pretrained (Part 1-2)
- Les defis identifies ici (rarete des donnees, label space enorme, evaluation non standardisee) motivent les contributions de Part 3

-> plan du chapitre: on part des fondamentaux du NER, on zoome sur les benchmarks biomed, puis sur le codage medical, l'entity linking, les approches zero-shot, les donnees synthetiques, et enfin l'evaluation

4.2 Named Entity Recognition

4.2.1 Sequence Labeling

NER = identifier et classifier les mentions d'entites dans un texte.

Formulation classique: sequence labeling avec schema BIO (Beginning, Inside, Outside):

- Chaque token recoit un label: B-TYPE, I-TYPE, ou O
- Variantes: BILOU (Beginning, Inside, Last, Outside, Unit) pour plus de precision

CRF (Lafferty et al., 2001):

- Conditional Random Fields: modele les dependances entre labels consecutifs
- Evite les sequences invalides (ex: I-MALADIE apres B-TRAITEMENT)
- Standard pre-deep learning pour le NER

BiLSTM-CRF (Lample et al., 2016):

- Combine BiLSTM pour les representations contextuelles + CRF pour le decodage
- SOTA pre-BERT sur la plupart des benchmarks NER
- Character-level embeddings pour capter la morphologie

4.2.2 BERT-based NER

Devlin et al. (2019): BERT + linear layer sur chaque token -> NER.

- Simple: pretrained encoder + classification head, fine-tuning end-to-end
- Pas besoin de features manuelles ni de CRF (debat: CRF aide parfois marginalement)
- Representations contextuelles profondes: "cancer" dans "cancer du poumon" vs "cancer du sein" -> vecteurs differents
- Transfer learning: le modele pretrained apporte des connaissances linguistiques generales

4.2.3 Au-dela du token-level

Span-based NER:

- Au lieu de labelliser chaque token, on enumere les spans candidats et on classifie chaque span
- Avantage: gere mieux les entites imbriquees (nested entities)
- Exemple biomed: "cancer du poumon droit" = [MALADIE [ANATOMIE]]
- Les entites imbriquees sont frequentes en biomed: symptomes dans des maladies, anatomie dans des procedures

-> ok le NER marche bien en general, mais le domaine biomedical a ses specificites: vocabulaire technique, entites imbriquees, negation, style telegraphique

4.3 Biomedical and Clinical NER

4.3.1 Benchmarks anglais

NER sur texte scientifique (PubMed):

- BC5CDR (Li et al., 2016): chemical-disease relations, 1500 articles PubMed, 2 types d'entites (chemical, disease)
- NCBI Disease (Dogan et al., 2014): mentions de maladies dans 793 abstracts PubMed
- JNLPBA (Collier & Kim, 2004): genes/proteines, corpus GENIA, 5 types d'entites
- AnatEM (Pyysalo & Ananiadou, 2014): entites anatomiques, 1212 documents
- ChemProt (Krallinger et al., 2017): interactions chemical-protein, relation extraction

NER sur texte clinique:

- i2b2 shared tasks (Uzuner et al., 2010, 2011): NER clinique sur notes MIMIC, plusieurs editions (medication extraction, coreference, relations temporelles)
- n2c2 (successeur de i2b2): n2c2 2018 (Henry et al., 2020) = adverse drug events + criteres d'eligibilite aux essais cliniques. n2c2 2022 = contextualization of clinical NLP
- Taches specifiques sur notes cliniques: de-identification (DEID), social determinants (SDoH), phenotyping

Classification:

- GAD (Bravo et al., 2015): gene-disease associations, classification binaire
- Hallmarks of Cancer (Baker et al., 2016): classification multi-label de hallmarks tumoraux

BLURB (Gu et al., 2022): Biomedical Language Understanding and Reasoning Benchmark, unifie 13 datasets biomed en un seul benchmark. PubMedBERT = premier modele evalue.

4.3.2 Benchmarks francais

QUAERO (Neveol et al., 2014):

- EMEA: notices medicamenteuses en francais
- MEDLINE: titres d'articles scientifiques
- 10 types d'entites suivant les semantic groups UMLS (Anatomy, Chemical, Device, Disorder, Gene, Living Being, Object, Phenomena, Physiology, Procedure)
- Entites imbriquees presentes
- Utilise dans CLEF eHealth 2016 (Neveol et al., 2016)

CAS / DEFT 2020 (Cardon et al., 2020):

- Cas cliniques issus d'articles scientifiques francais
- Tache 3 de DEFT 2020 = extraction d'information clinique
- CAS1: 2 classes (pathologie, signes/symptomes)
- CAS2: 8 classes (anatomie, dose, examen, mode, moment, substance, traitement, valeur)
- Devenu benchmark standard pour le NER clinique francais

E3C (Magnini et al., 2021):

- Corpus multilingue (EN, FR, ES, IT, EU)
- Cas cliniques, notices medicamenteuses, resumes de these
- Layer 1 = annote manuellement, layer 2 = semi-annote, layer 3 = non annote
- 1 seule classe: entite clinique (pas de sous-types)

CLEF eHealth 2016 (Neveol et al., 2016):

- Task 2: NER + normalisation sur corpus QUAERO
- Codage CIM-10 sur certificats de deces (corpus CepiDC)
- BRATeval = outil d'evaluation officiel (exact match sur character offsets)
- 7 equipes participantes

DrBenchmark (Labrak et al., 2024, LREC-COLING):

- Premier benchmark unifie pour le francais biomedical (equivalent de BLURB pour l'anglais)
- 20 taches sur 12 datasets: NER (9 taches), classification (8), QA (1), similarite semantique (2)
- Regroupe QUAERO, E3C, CAS/DEFT-2020, DEFT-2021, Mantra-GSC, FrenchMedMCQA, PxCorpus, MorFITT, CLISTER, DiaMed
- Permet enfin une comparaison equitable entre modeles sur le francais biomedical
- Resultat cle: les modeles specialises biomed surpassent les modeles generaux

4.3.3 Specificites du texte clinique

Le texte clinique est tres different du texte scientifique (PubMed) et du texte general (web):

- Style telegraphique: phrases incompletes, listes de symptomes, abreviations non standard
- Abreviations: "IDM" (infarctus du myocarde), "HTA" (hypertension arterielle), "NFS" (numeration formule sanguine)
- Negation: "pas de signe de pneumonie" -> "pneumonie" est mentionnee mais absente. NegEx (Chapman et al., 2001): algorithme a base de regles pour detecter la negation, toujours utilise comme baseline
- Temporalite: antecedents vs diagnostics actuels vs hypotheses
- Incertitude: "probable", "a surveiller", "a eliminer"
- Fautes d'orthographe frequentes (saisie rapide par les cliniciens)

Consequence: les modeles entraines sur du texte scientifique (PubMed) ou general (web) sont sous-optimaux pour le texte clinique -> d'ou les modeles pretrained sur le domaine (BioBERT, PubMedBERT, ClinicalBERT, GatorTron, DrBERT, cf Section 2.7). Ces modeles specialises ameliorent les performances sur les taches biomed, mais restent supervises: il faut annoter des exemples pour chaque classe cible.

Alternative architecturale: Clinical Mamba (Yang et al., 2024) adapte les State Space Models (SSM) au domaine clinique. Complexite lineaire en la longueur de sequence (vs quadratique pour les transformers) -> pertinent pour les documents cliniques longs. Premiers resultats competitifs avec les transformers.

-> on a les benchmarks et les modeles pour le NER biomed, mais le codage medical c'est pas juste du NER: c'est assigner des codes d'une ontologie a un document entier, avec des milliers de labels possibles

4.4 Medical Coding

4.4.1 Le probleme

Medical coding = assigner des codes standardises (CIM-10, CCAM, ATC) a un document clinique.

En France:

- PMSI: chaque sejour hospitalier doit etre code
- DP (Diagnostic Principal), DAS (Diagnostics Associes), actes CCAM
- 30M sejours/an (ATIH), code manuellement par des codeurs professionnels
- Erreurs de codage = impact financier et epidemiologique

Defis:

- Label space enorme: 17K codes CIM-10 (ATIH), 8K codes CCAM, 6K codes ATC
- Documents longs: comptes-rendus de sortie = souvent 1000-3000 tokens, depassent la limite de BERT (512)
- Distribution tres desequilibree: quelques codes frequents (I10 Hypertension, E11 Diabete), longue traine de codes rares
- Multi-label: un sejour peut avoir 1 DP + plusieurs DAS

4.4.2 Approches

CAML (Mullenbach et al., 2018, NAACL):

- Convolutional Attention for Multi-Label classification
- CNN + label-wise attention: chaque code a son propre mecanisme d'attention sur le document
- Premiere approche explainable pour ICD coding: on peut visualiser quels passages du texte ont motive chaque code
- DR-CAML: regularisation par les descriptions textuelles des codes

HyperCore (Cao et al., 2020, ACL):

- Exploite la hierarchie ICD via embeddings hyperboliques (Poincare ball)
- L'espace hyperbolique capture naturellement les structures arborescentes (distance exponentielle avec la profondeur)
- + graphe de co-occurrence entre codes (GCN)

PLM-ICD (Huang et al., 2022, ClinicalNLP@NAACL):

- Premier a utiliser des pretrained language models pour ICD coding
- Label attention sur les representations du PLM
- Segment pooling pour gerer les documents longs (decoupe en segments, pool les representations)
- Resultat cle: meilleur encoder pretrained = meilleur codage -> la qualite du pretraining impacte directement le codage

MSMN (Yuan et al., 2022, ACL):

- Multiple Synonyms Matching Network
- Exploite les synonymes UMLS pour chaque code ICD
- Le texte clinique utilise des expressions variees pour un meme concept: "infarctus du myocarde", "IDM", "crise cardiaque" -> tous I21
- Multi-synonym matching: representations de codes plus riches que la seule description officielle

Et les LLMs? Soroush et al. (2024, NEJM AI):

- Evaluation de GPT-4 pour le codage ICD-10: exact match = 33.9% seulement

- Les LLMs generatifs peinent sur le codage: label space trop grand, codes trop specifiques
- Les approches specialisees (PLM-ICD, MSMN) restent superieures
- -> renforce le besoin d'encoders specialises plutot que de LLMs generiques

4.4.3 Codage en francais

Ressources specifiques:

- FRACCO (Pignat et al., 2024): corpus d'oncologie francais, 1301 cas cliniques annotes avec codes CIM-O-3.1, classification multi-label sur 100 codes
- FRASIMED (Zaghir et al., 2023): adaptation francaise de Cantemist (codage tumeurs, Miranda et al. 2020) et Distemist (codage maladies, Miranda et al. 2022), 2051 documents cliniques traduits de l'espagnol
- CepiDC / CLEF eHealth 2016: codage CIM-10 sur certificats de deces francais

-> le codage marche sur MIMIC (anglais), mais: 17K labels = impossible d'annoter des exemples pour chacun. En francais, les ressources sont encore plus rares. -> besoin d'approches qui generalisent a des labels non vus = zero-shot. Mais d'abord: comment associer mentions et concepts?

4.5 Entity Linking and Normalization

4.5.1 Le probleme

Entity linking = associer une mention textuelle a une entite dans une base de connaissances (UMLS, SNOMED-CT, CIM-10).

Normalization = mapper des variantes terminologiques au concept canonique:

- "infarctus du myocarde", "IDM", "crise cardiaque" -> I21.9 (CIM-10)
- "diabete de type 2", "DNID", "diabete non insulinodependant" -> E11 (CIM-10)

Difference avec le codage medical:

- Entity linking = mention-level: associer chaque mention a un concept
- Medical coding = document-level: assigner des codes a un document entier
- Entity linking peut etre vu comme une brique du codage: d'abord extraire les mentions (NER), puis les normaliser (linking), puis agreger au niveau document

4.5.2 Approches

SapBERT (Liu et al., 2021, NAACL):

- Self-Alignment Pretraining for BERT
- Contrastive learning sur synonymes UMLS: 4M+ concepts
- Synonymes = positifs, non-synonymes = negatifs
- SOTA entity linking biomed sur 6 benchmarks
- "One-model-for-all": un seul modele pour tous types d'entites (maladies, medicaments, procedures)
- Base: PubMedBERT fine-tune avec metric learning

CODER (Yuan et al., 2022):

- Contrastive pretraining sur 87M triplets de relations UMLS
- Ajoute les relations (is-a, part-of, causes) en plus des synonymes dans l'objectif contrastif
- Medical term normalization

BioSyn (Sung et al., 2020, ACL):

- Bi-encoder + cross-encoder pour entity normalization
- Marginal score aggregation: combine scores de plusieurs synonymes
- Montre que le re-ranking avec cross-encoder ameliore significativement

4.5.3 Limites

- La plupart des travaux sont en anglais et sur UMLS
- UMLS couvre bien l'anglais mais pas le francais clinique (abreviations, synonymes informels)
- SapBERT et CODER ne captent pas la hierarchie complete de l'ontologie (parents, exclusions, complications)
- Pas d'exploitation des relations RDF/SKOS (broader, narrower, related, exactMatch)

-> on sait normaliser des mentions isolees, mais comment extraire ET normaliser sans supervision, pour des milliers de labels jamais vus a l'entrainement?

4.6 Zero-shot and Open NER

4.6.1 Pourquoi zero-shot

- Les labels medicaux evoluent: nouveaux codes ICD, nouveaux criteres d'essais cliniques, nouvelles maladies (COVID-19)
- Impossible d'annoter des exemples pour chaque label (17K codes CIM-10)
- Les modeles NER classiques (BERT + linear head) ont un nombre fixe de classes de sortie
- -> besoin de modeles qui generalisent a des types d'entites non vus a l'entrainement

4.6.2 GLiNER

GLiNER (Zaratiana et al., 2024, NAACL):

- Formulation: NER = matching entre spans de texte et descriptions de types d'entites
- Architecture: un seul transformer bidirectionnel encode conjointement le texte et les types d'entites
- Les types d'entites sont donnes en langage naturel (pas des labels fixes)
- Token [ENT] devant chaque type -> representation du type
- Span representation layer -> representation de chaque span candidat
- Matching par produit scalaire + sigmoid dans un espace partage
- Bat ChatGPT sur zero-shot NER tout en etant beaucoup plus petit (200M params)

GLiNER2 (Zaratiana et al., 2025, EMNLP):

- Extension de GLiNER a un framework multi-taches unifie: NER, relation extraction, classification, extraction structuree
- Meme architecture de base mais avec des tokens de relation en plus des tokens d'entite
- -> un seul modele pour plusieurs taches IE, toujours en zero-shot

GLiNER-BioMed (Zaratiana et al., 2025):

- Adaptation de GLiNER au domaine biomedical

- Fine-tuning sur des datasets NER biomed (BC5CDR, NCBI, JNLPBA, etc.)
- +5.96% F1 par rapport au GLiNER general sur les benchmarks biomed
- Montre que l'adaptation au domaine est necessaire meme pour les modeles zero-shot

-> GLiNER et ses extensions montrent que le matching span-label est une direction prometteuse. Mais ou trouver les donnees d'entrainement?

Pre-entrainement sur Pile-NER:

- UniversalNER (Zhou et al., 2024, ICLR): distillation de ChatGPT
- ChatGPT annote des passages du Pile -> 45K paires, 13K types d'entites distincts
- Instruction-tuning de LLaMA sur ces donnees
- Bat ChatGPT de 7-9 F1 points en moyenne sur NER
- GLiNER pre-entraine sur ce dataset = forte capacite zero-shot

4.6.3 Autres approches

GoLLIE (Sainz et al., 2024, ICLR):

- Guideline-following Large Language model for IE
- Approche generative: le LLM genere des annotations structurees a partir de guidelines en langage naturel
- Zero-shot: les guidelines decrivent les types d'entites -> pas besoin d'exemples annotes
- Avantage: peut suivre des definitions complexes (criteres d'inclusion, codes CIM avec exclusions)
- Mais: modele genératif = plus lent et plus gros qu'un encoder

NuNER (Bogdanov et al., 2024, EMNLP):

- Foundation model NER compact (RoBERTa-base, 125M params)
- Pre-entraine sur annotations GPT-3.5 de C4: 24.4M mots, 4.38M annotations, 200K types d'entites
- Contrastive learning pour specialiser l'encoder
- Resultat cle: taille ET diversite du dataset d'entrainement = facteurs critiques

LLMs pour clinical NER:

- Agrawal et al. (2022): LLMs pour clinical IE, performances prometteuses
- Singhal et al. (2022): Med-PaLM, QA medicale a l'echelle
- Mais: Lehman et al. (2023), "We May Have Overestimated LLMs in Clinical NLP"
  - BERT-type models restent competitifs pour les taches extractives cliniques
  - LLMs trop gros pour deploiement hospitalier (contraintes infra)
  - Problemes de confidentialite avec APIs proprietaires (GPT-4, Claude)
  - -> les encoders restent pertinents pour les taches extractives cliniques

4.6.4 Limites du zero-shot actuel

- GLiNER pre-entraine sur donnees generales (Pile-NER), pas cliniques
- Les types d'entites sont des descriptions textuelles courtes: ne capte pas la structure hierarchique des ontologies
- Pas d'exploitation des relations entre codes (parent, exclusion, synonyme)
- Performances degradees sur les codes rares (longue traine)

-> zero-shot NER prometteur, mais: GLiNER ignore la structure ontologique, et les donnees d'entrainement sont generales. Les donnees cliniques sont privees -> comment entrainer sur du domaine clinique sans donnees reelles?

4.7 Synthetic Data for Clinical NLP

4.7.1 Le probleme de la rarete

- Donnees cliniques = privees (CNIL en France, HIPAA aux US)
- Modeles entraines sur donnees hospitaliers = non partageables entre etablissements
- Peu de datasets annotes publics en francais clinique: QUAERO, CAS, E3C = petits et principalement scientifiques, pas cliniques
- Les datasets cliniques accessibles (MIMIC) = anglais, un seul hopital americain

4.7.2 LLM-as-annotator

Paradigme recent: utiliser un LLM pour annoter des donnees, puis distiller vers un petit modele:

- FineWeb-Edu (Penedo et al., 2024): Llama-3-70B annote 500K docs, distille vers classifieur BERT-like (cf Ch.2)
- UniversalNER (Zhou et al., 2024): ChatGPT annote le Pile pour NER, distille vers LLaMA
- NuNER (Bogdanov et al., 2024): GPT-3.5 annote C4 pour NER, distille vers RoBERTa
- Avantage: le LLM capture des connaissances medicales, le petit modele est deployable en hopital

4.7.3 Generation de notes cliniques synthetiques

Premiers travaux: generer directement des notes cliniques synthetiques avec des modeles de langue.

Ive et al. (2020, npj Digital Medicine):

- Generation de notes cliniques synthetiques en psychiatrie (discharge summaries)
- Objectif: contourner la confidentialite des donnees
- Resultat: les modeles NLP entraines sur donnees synthetiques atteignent des performances comparables a ceux entraines sur donnees reelles
- Premier a valider experimentalement que le synthetique peut remplacer le reel pour les taches NLP cliniques

Chintagunta et al. (2021, NLPMC@ACL):

- GPT-3 pour augmenter des donnees de dialogue medical: 210 exemples annotes -> performances equivalentes a 6400 exemples (30x)
- Low-shot + ensemble: le LLM genere des variantes, un ensemble filtre les generations de mauvaise qualite
- Montre que le LLM-as-augmenter fonctionne specifiquement en domaine medical

Asclepius (Kweon et al., 2024, ACL Findings):

- LLM clinique entierement entraine sur notes synthetiques

- Pipeline: case reports publics (PMC) -> GPT-3.5 genere des notes cliniques realistes a partir des cas
- Aucune donnee clinique privee utilisee
- Performances comparables aux modeles entraines sur MIMIC (donnees reelles)
- -> preuve que des notes synthetiques de qualite suffisent pour entrainer des modeles cliniques

Approche texte + annotations simultanees:

- Le LLM genere le texte ET les annotations en meme temps
- Avantage: alignement texte-annotation garanti (pas de desynchronisation)
- Inputs structurees possibles: informations patient, codes CIM-10, guidelines therapeutiques

Evaluation de la qualite:

- Evaluation aveugle par medecins: notes synthetiques vs reelles
- Criteres: coherence medicale, completude, qualite linguistique
- Observation recurrente: les notes synthetiques sont souvent plus completes mais moins coherentes medicalement que les notes reelles

Risques:

- Style different des notes reelles (trop propre, pas assez telegraphique)
- Biais du LLM: peut generer des cas stereotypes
- Les annotations LLM ne sont pas gold-standard: heuristiques neurales, pas expertises medicales

-> les donnees synthetiques sont une solution a la rarete, mais comment evaluer correctement les modeles entraines dessus? Et plus generalement, comment comparer equitablement des modeles avec des tokenizers differents?

4.8 Evaluation

4.8.1 Metriques NER

seqeval (strict entity-level):

- Evaluation au niveau entite: une entite est correcte si et seulement si le type ET les frontieres sont exacts

- Schema IOB2: B-TYPE marque le debut, I-TYPE la continuation, O l'absence d'entite
- Micro-average: aggrege toutes les entites (favorise les classes frequentes)
- Macro-average: moyenne des scores par classe (traite toutes les classes egalement)
- Standard de facto pour le NER en NLP

BRATeval (CLEF eHealth 2016):

- Exact match sur character offsets dans le texte original
- Avantage: completement independant du preprocessing et de la tokenisation
- Utilise comme outil officiel pour les campagnes CLEF eHealth
- Mais: moins repandu que seqeval dans la communaute NLP

Token-level classification (weighted F1):

- Chaque token recoit un label, evaluation token par token
- Probleme majeur: le token "O" (non-entite) est largement majoritaire
- Le weighted F1 est domine par la classe O -> scores artificiellement eleves
- Utilise par DrBERT (Labrak et al., 2023): rend la comparaison avec d'autres modeles difficile

4.8.2 Impact de la tokenisation

- Tokenizers differents decoupent le texte differemment
- SciBERT (Beltagy et al., 2019): 42% intersection avec le vocabulaire BERT
- Un tokenizer qui sur-segmente les termes techniques = plus de tokens a predire = plus de chances d'erreur
- Consequence: les metriques token-level sont biaisees par le choix du tokenizer
- Exemple: "echocardiographie" = 3 tokens (CamemBERT) vs 2 tokens (tokenizer specialise) -> nombre de predictions different -> score different

4.8.3 Besoin d'un benchmark unifie

Problemes actuels:

- Chaque papier utilise des datasets, metriques, splits, et hyperparametres differents
- Pas de protocole standard pour le francais biomedical
- CRF vs linear head, gel des couches vs fine-tuning complet, micro vs macro F1
- Comparaison entre modeles = difficile et souvent biaisee

Comparaison equitable entre architectures:

- BERT-type (encoders) vs LLMs (decoders) vs GLiNER (bi-encoder): architectures tres differentes
- Tokenizers differents, procedures de fine-tuning differentes
- -> besoin d'une evaluation au niveau caractere, independante du tokenizer, qui permette une comparaison equitable

4.9 Limites et transition

- Rarete des donnees cliniques: les textes hospitaliers sont prives (CNIL), les modeles entraines dessus non partageables. Les corpus publics francais (QUAERO, CAS, E3C) sont petits et principalement scientifiques -> besoin d'alternatives aux donnees hospitalieres
- Label space enorme: 17K codes CIM-10, 8K CCAM, impossible d'annoter des exemples pour chaque code. Les approches supervisees classiques (BERT + linear head) ne scalent pas -> besoin d'approches zero-shot capables de generaliser a des labels non vus
- Zero-shot NER existant = pre-entraine sur donnees generales: GLiNER et UniversalNER sont entraines sur Pile-NER (texte web general), pas sur du domaine clinique. Performances degradees sur les codes rares -> besoin d'adapter au domaine medical
- Ontologies sous-exploitees dans le NER: GLiNER utilise des descriptions textuelles comme types d'entites mais ignore la structure hierarchique, les exclusions, les synonymes encodes dans RDF/SKOS -> besoin d'integrer la structure ontologique dans les embeddings de labels

- Evaluation non standardisee: pas de benchmark unifie pour le francais biomed, metriques biaisees par le tokenizer, protocoles incomparables entre papiers -> besoin d'un framework d'evaluation tokenizer-agnostic
- Encoders vs decoders: malgre l'engouement pour les LLMs, les encoders restent competitifs pour les taches extractives cliniques (Lehman et al., 2023) et sont deployables sur les infrastructures hospitalieres

-> ce chapitre a couvert les taches de clinical IE: NER, codage medical, normalisation, et les defis specifiques du domaine clinique. Les limites identifiees (rarete, label space, zero-shot non adapte, ontologies sous-exploitees, evaluation non standardisee) definissent les axes de recherche de la partie suivante.

Part II

Building a Biomedical Corpus

5 Collecting Biomedical Text

5.1 Introduction

Hospitals produce millions of clinical documents every year. These can be used to train language models internally, but the resulting models inherit the legal constraints of their training data: they cannot be released publicly or exchanged between institutions. For a model that is truly open, the training data must come from somewhere else.

Yet hospital records remain the richest source of medical information. Clinical data warehouses have made them increasingly accessible for research purposes. It is estimated that up to 80% of entities are missing from structured modalities (Raghavan et al., 2014), making clinical reports the richest source of medical information. Yet extracting this information requires high-performing language models adapted to the biomedical domain.

BERT-based models (Devlin et al., 2019a) consistently demonstrate state-of-the-art results for a wide range of natural language processing tasks. The adaptation of BERT to the French language, particularly with the CamemBERT model (Martin et al., 2020a), has replicated these performances in French natural language processing. However, the results of this model on biomedical data are disappointing (Cardon et al., 2020), as it exhibits lower performance compared to heuristic models on certain evaluation datasets. These results are predictable because CamemBERT is trained on plain language, often sourced from web pages such as forums. Biomedical data, especially clinical data, are significantly different. They contain technical terms that are very rare or absent in everyday language, and they have a radically distinct style, often telegraphic, rarely consisting of complete sentences, with varying abbreviations.

In this chapter, we address this gap by introducing *biomed-fr*, a public French biomedical corpus of 413 million words assembled from three complementary sources. We describe the corpus construction, analyze the vocabulary mismatch between general and biomedical French tokenizers, and situate our effort among existing French biomedical corpora. The use of this corpus for continual pretraining of CamemBERT, yielding CamemBERT-bio, is presented in Chapter 8.

Authors	Source	Size	Public
Copara et al. (2020)	PubMed	31K docs	Yes
Le Clercq de Lannoy et al. (2022)	Misc.	136M words	Partial
Dura et al. (2022)	APHP CDW	21M docs	No
Labrak et al. (2023)	Misc.	7.4 GB	Partial

Table 5.1: French biomedical corpora used for language model adaptation. Size units vary across works as reported by their respective authors.

5.2 Existing French Biomedical Corpora

Several works have attempted to adapt CamemBERT to the biomedical domain, each relying on different training data. Table 5.1 summarizes these corpora.

Copara et al. (2020) explored a second phase of pre-training on 31,000 French biomedical scientific articles. However, they did not observe a significant improvement on a clinical named entity recognition task using the large version of CamemBERT. This could be explained by the combination of a relatively small corpus (31K documents compared to the 18 billion words in BioBERT) and the large version of the CamemBERT model.

Le Clercq de Lannoy et al. (2022) also adapted CamemBERT to the biomedical domain. They aggregated documents from various sources, including PubMed, Cochrane, ISTEX, and Wikipedia, forming a larger partially public corpus of approximately 136 million words. They observed a 2-point improvement in F-measure on a named entity recognition evaluation set composed of drug notices (EMEA), but no significant improvement on a set composed of scientific article titles (MEDLINE).

Dura et al. (2022) continued the pre-training of CamemBERT on 21 million clinical documents from the APHP (Assistance Publique – Hôpitaux de Paris) clinical data warehouse. They observed a significant 3% improvement on APMed, a private clinical named entity recognition dataset owned by APHP. They also achieved similar scores to CamemBERT on EMEA and MEDLINE. Their new model performs better on clinical data, yet it obtains scores similar to CamemBERT in other biomedical domains.

Labrak et al. (2023) introduced a public French biomedical model named DrBERT. Through their experiments, they explored both continual-pretraining and from-scratch training strategies. Their findings indicated a superior performance when training from scratch, suggesting that continual-pretraining with CamemBERT for French biomedical data may not be as effective. However, when applied to PubMedBERT, continual-pretraining yielded results nearly on par with the from-scratch approach.

Finally, Berhe et al. (2023) introduced AliBERT, which was trained on a French biomedical corpus primarily comprising articles from ScienceDirect and theses collected through Sudoc. The model leverages a new regularized Unigram-based

Source	Size (M words)	Content
ISTEX	276	Scientific articles
CLEAR	73	Drug leaflets
E3C	64	Clinical cases, leaflets
Total	**413**	

Table 5.2: Composition of the *biomed-fr* corpus (in millions of words).

tokenizer and underwent extensive training on 48 GPUs for a total duration of 20 hours. Their pretrained model outperforms notable French non-domain-specific models, such as CamemBERT and FlauBERT, in two biomedical downstream tasks. Unfortunately, the model is not currently available.

5.3 The biomed-fr Corpus

We built a French biomedical corpus composed exclusively of public documents to minimize the usage constraints mentioned earlier. The documents come from three different sources (see Table 5.2), the main one being ISTEX. This new corpus, named *biomed-fr*, consists of 413 million words, equivalent to 2.7 GB of data. Martin et al. (2020a) have shown that with only 4 GB of data, it is possible to achieve performance almost comparable to the model trained with the 138 GB OSCAR dataset (Ortiz Suárez et al., 2019). For an adaptation of CamemBERT to the biomedical domain, this amount of data can be considered sufficient.

ISTEX. The ISTEX database contains references to 27 million scientific publications. We extracted 108,183 French documents published in a biology or medical journal since 1990. Articles published before this date often contain numerous typographical errors, as they are often scanned articles that require optical character recognition algorithms, resulting in a certain number of errors. Such errors are found to a lesser extent in articles published after 1990. Some documents, although in French, contain passages in English. Therefore, there is an indeterminate amount of English in this corpus. However, it is unlikely that this will significantly impact pre-training. Typographical errors and the presence of other languages are aspects that can be addressed in future versions of *biomed-fr*.

CLEAR. The CLEAR corpus (Grabar and Cardon, 2018) consists of encyclopedia articles, drug leaflets, and abstracts of scientific articles. Each document is available in two versions: one in technical language and the other in simplified language. We retrieved all of these documents in both versions. Regarding the drug leaflets, we removed redundant sentences at the beginning and end of each document, such as

Term	General	Specialized
échocardiographie	écho-cardi-ographie	échocardiographi-e
transthoracique	trans-thorac-ique	trans-thoracique
glimépiride	g-lim-épi-ride	gli-m-épi-ride
cardiopathie	cardio-pathie	cardiopathie
diastoliques	dia-s-tol-iques	diastolique-s

Table 5.3: Comparison of tokenization between a general-purpose tokenizer and a specialized tokenizer for some biomedical technical terms. Hyphens indicate subword boundaries.

the website navigation bar from which the documents were extracted or information about the company selling the documents.

E3C. This corpus (Magnini et al., 2020) is composed of three layers. The first two layers are annotated or semi-annotated and will be used for evaluation in Chapter 8. The last layer is not annotated, and that is the one we retrieved for *biomed-fr*. It consists of medical specialty admission competitions, drug leaflets, and medical thesis abstracts. There may be duplicates of some leaflets found in the CLEAR corpus.

BIOMED-FR-SMALL. By randomly selecting 10% of content from *biomed-fr*, we created a smaller corpus called *biomed-fr-small*. This subset allows us to study the impact of corpus size on downstream performance. Results of this comparison are presented in Chapter 8.

5.4 Vocabulary Analysis

CamemBERT-bio is a biomedical-adapted model based on CamemBERT. Unlike a new model trained from scratch, it shares the same vocabulary. The vocabulary of CamemBERT was constructed using SentencePiece (Kudo and Richardson, 2018) on an OSCAR sample. Therefore, it is a general-purpose vocabulary designed for everyday language. We can hypothesize that the tokenization of CamemBERT may result in oversplitting of biomedical technical terms.

To investigate this possibility, we trained a specialized tokenizer on *biomed-fr-small* and calculated the intersection of the two vocabularies.

We find a 45% intersection between the two vocabularies (Table 5.3), which is quite close to the 42% intersection found by Beltagy et al. (2019) between the vocabulary of BERT and SciBERT. Therefore, there is a significant difference in the most frequent terms.

The relatively modest performance gain of PubMedBERT compared to BioBERT and the similar performance of DrBERT compared to CamemBERT-bio despite their specialized vocabulary, together with the over-segmentation of technical terms and the low intersection rate, demonstrate the value of investigating vocabulary adaptation. However, taking into account the comparable performance levels and the significant difference in computational cost (see Chapter 8), we advocate for the continual-pretraining method as the preferred adaptation approach.

5.5 Limitations

It is important to discuss some limitations of our corpus.

Firstly, *biomed-fr* is limited in its diversity. This is because we chose to only use publicly available materials, which tend to lean towards scientific content. As a result, models trained on this corpus may not fully represent performances on real private clinical data. The gap between scientific and clinical text is a recurring challenge that we address in Chapter 6.

Furthermore, we did not explore any potential biases in our data and some parts of the dataset would benefit from further cleaning. These aspects could impact the outputs of models trained on this corpus and should be investigated.

Third, all documents from the selected sources were included indiscriminately, without assessing whether individual passages were genuinely useful for pretraining. Whether such filtering would improve downstream performance is an open question that we explore in Chapters 6 and 7.

Conclusion

More than half of the most frequent biomedical subwords are absent from CamemBERT's general vocabulary. A mismatch this large raises a natural question: should we abandon the general tokenizer entirely and train a new model from scratch? The answer, perhaps surprisingly, is no, as we show in Chapter 8.

Before we get there, however, a more immediate question deserves attention. We included all 413 million words of *biomed-fr* without filtering. Not all of this text is equally useful. In the next chapter, we investigate whether selecting data by quality can improve upon including everything indiscriminately.

6 Detecting Content Types

6.1 Introduction

For web data, the question of what to keep and what to discard has a straightforward answer. FineWeb-Edu (Penedo et al., 2024) showed that filtering entire documents by educational quality, discarding 92% of the data, actually improves downstream performance. A low-quality web page is low-quality throughout: if the source is bad, every paragraph is bad. But a scientific article is not a web page.

Figure 6.1 illustrates the difference. A web page about astronomy scores low throughout and can be safely discarded. A PMC article on pulmonary hypertension, by contrast, contains a textbook-quality review paragraph, a boilerplate committee note, and a clinical case report with real diagnostic values. Document-level filtering would either keep all three or discard all three. Neither is the right answer.

In the previous chapter, we assembled *biomed-fr* by including all available documents from three public sources. This approach treats every paragraph equally. How much of this text actually helps, what types of content does it contain, and can we be smarter about what we include?

In this chapter, we introduce a paragraph-level annotation framework that distinguishes content types, domains, and educational quality within scientific articles. We show that selective pretraining on the most relevant paragraphs matches the performance of training on the full corpus using one-third of the tokens. We also extract 2 million clinical case paragraphs from PubMed Central, providing an openly available alternative to restricted hospital records. Finally, we investigate the risks of neural quality filtering: not all selection strategies are equally safe, and some can amplify benchmark contamination by a factor of 20.

6.2 Content Heterogeneity in Scientific Articles

Large language models for biomedicine are typically pretrained on PubMed Central (PMC) Open Access articles. While valuable, this corpus presents challenges including heterogeneous quality, predominance of English content (approximately 98%), uneven representation of clinical cases, and variable educational value. Existing filtering approaches operate at the article level: BioMistral (Labrak et al., 2024) upsampled non-English articles, while Meditron (Chen et al., 2023) scored articles

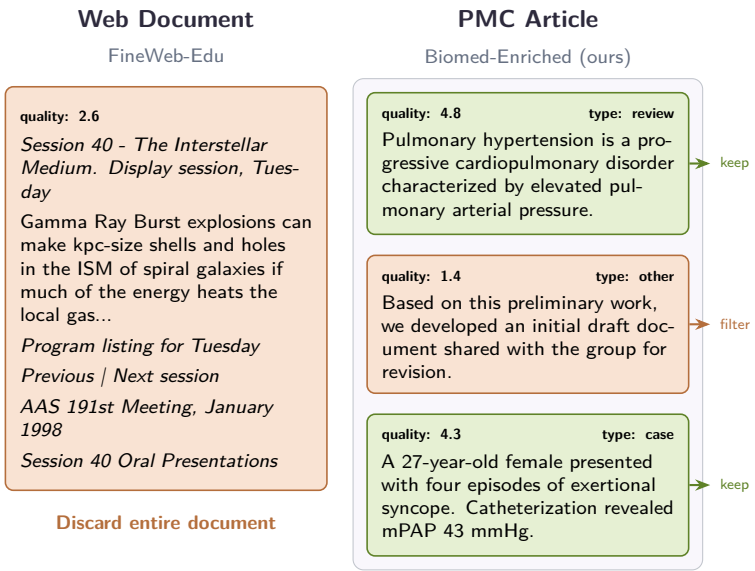

Figure 6.1: **Document-level vs. paragraph-level filtering.** A low-quality web page can be discarded entirely (left). A scientific article mixes high-value content (clinical cases, reviews) with boilerplate (right). Paragraph-level annotation allows keeping the valuable parts.

using MeSH tags, publication type, journal reputation, recency, and citation count. These methods assume that entire documents are either valuable or not.

This assumption does not hold for curated scientific corpora. Unlike web-scraped data where low-quality sources justify discarding entire documents, scientific articles are all potentially valuable. The real issue is variance *within* articles. To quantify this, we analyzed 10,000 PMC articles (378,000 paragraphs) and found that 91.6% of articles contain multiple document types (clinical case, study, review, other). Educational scores vary by 2.6 points on average within articles, spanning more than half of the 1–5 scale. Some paragraphs contain excellent educational or clinical content, while others add little domain knowledge but remain difficult to detect with simple heuristics.

6.3 Method

6.3.1 Data Collection and Preprocessing

We extracted text from the PubMed Central (PMC) Open Access Subset, containing approximately 4.5 million full-text scientific articles. Using a custom pipeline, we processed the raw XML files to extract article content, segment articles into 133 million individual paragraphs, filter out non-textual elements, and retain only paragraphs containing a minimum of 64 tokens.

	Single	Mixed
Document Type	8.4%	91.6%
Domain	44.7%	55.3%
Educational Score (1–5 scale)		
	Mean	**Median**
Intra-article std	0.68	0.70
Intra-article range	2.61	2.86

Table 6.1: Intra-document heterogeneity in 10K PMC articles (378K paragraphs). 91.6% of articles contain multiple document types. Educational scores vary by 2.6 points on average within articles.

Dimension	Agreement
Educational score	$r = 0.876$, MAE $= 0.326$
Domain	$F_1 = 0.954$
Document type	$F_1 = 0.925$, $\kappa = 0.821$

Table 6.2: Distillation quality: agreement between the XLM-RoBERTa student and Llama-3.1-70B teacher on held-out paragraphs.

6.3.2 Two-Stage Annotation Framework

We follow a two-stage annotation process on a single 8×A100 node: 160 GPU-hours for the first stage and 80 GPU-hours for the second.

Stage 1: LLM annotation. We used Llama-3.1-70B-Instruct (Grattafiori et al., 2024) to annotate a diverse subset of 400,000 paragraphs (approximately 0.33% of the corpus) across multiple dimensions: type classification (clinical case, study, review, other), domain categorization (clinical, biomedical, other), and educational quality (1–5 scale). Language was identified separately using langdetect.

Stage 2: Distillation. To scale annotation to the full corpus, we distilled the LLM annotations into a smaller XLM-RoBERTa-base model (Conneau et al., 2020) trained to jointly predict all annotation dimensions. On 39,800 held-out paragraphs, the student closely matches the Llama-3.1-70B labels (Table 6.2). It is important to note that these annotations are not gold-standard labels verified by medical experts, but neural heuristics for data curation. Similar to FineWeb-Edu (Penedo et al., 2024), they guide which content to prioritize during pretraining.

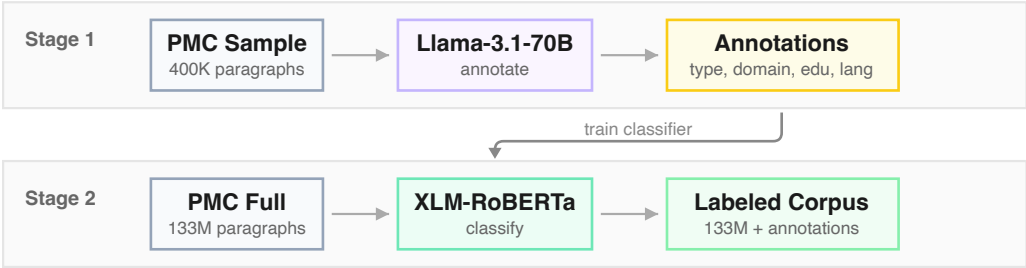

Figure 6.2: **Two-stage annotation pipeline.** Stage 1: Llama-3.1-70B annotates 0.33% of PMC paragraphs across four dimensions. Stage 2: a fine-tuned XLM-RoBERTa distills these annotations to label the full corpus, enabling flexible filtering strategies.

Figure 6.3: **Distribution of educational quality scores by document type and domain.** Reviews and studies show the highest proportion of high scores, while clinical texts display more variance. Content labeled "other" rarely reaches high scores.

6.3.3 Data Analysis

Most PMC paragraphs receive an educational score of 4, with a mean of 3.48 and median of 4.00. This distribution varies by content type: reviews and studies contain the highest proportion of educational content (86.9% and 78.7% scoring 4, respectively), while clinical cases show more variance (57.0% rated 4). Domain-wise, biomedical paragraphs score higher (75.3% at score 4) than clinical text (44.0%), and content labeled "other" rarely reaches high scores (2.1%). These patterns support the score threshold (≥ 3) used in BE-Educational and BE-All, and explain why combining domain and quality filters yields consistent gains across tasks.

Variant	Strategy
BE-Base	Complete unmodified PMC Open Access Subset (baseline).
BE-Educational	Removes paragraphs with educational quality scores below 3.
BE-Clinical	Upsamples 10× articles with predominantly clinical domain content.
BE-ClinicalCase	Upsamples 10× articles containing at least one clinical case paragraph.
BE-French	Upsamples 10× articles containing French text.
BE-All	Combines quality filtering (score ≥ 3), upsampling of clinical content, French text, and clinical cases, plus metadata prefixing.

Table 6.3: Biomed-Enriched dataset variants. All variants preserve the original article structure and use an 8K context window during pre-training.

6.3.4 Dataset Construction

Using the distilled model, we annotated the full PMC corpus and constructed several dataset variants through strategic filtering and upsampling, summarized in Table 6.3. Upsampling refers to duplicating articles in the training corpus, proportionally increasing their sampling probability.

For all variants, we preserved the original structure of the article and employed an 8K context window during pre-training. This ensures that models can process complete scientific articles, capturing dependencies where information presented in earlier paragraphs is essential for understanding later content.

6.4 Experimental Setup

Continual pre-training served as a method to evaluate the relevance and utility of our annotations. Our evaluation focuses on isolating the effects of data curation rather than pursuing state-of-the-art scores on benchmarks. A more powerful foundation model would likely yield higher absolute scores, but would probably obscure the precise impact of our dataset.

6.4.1 Decoder Setup

We selected OLMo2-7B-stage1 (Team et al., 2025) as our foundation model for continual pre-training, strategically choosing this intermediate checkpoint to more

Parameter	Value
Base model	OLMo2-7B-stage1
Peak LR	6.15e-5
Min LR	6.15e-6
LR Decay	Linear
Batch size	1024
Weight decay	0.1
Context	8,192 tokens
Hardware	128 MI250X
Tokens	33.6B / variant

(a) Decoder

Parameter	Value
Base model	ModernBERT-base
Peak LR	5e-4
Min LR	5e-7
LR Decay	one_minus_sqrt
Batch size	576
Weight decay	1e-5
Context	8,192 tokens
Hardware	4× H100
Tokens	70B

(b) Encoder

Table 6.4: Hyperparameters for continual pre-training.

clearly attribute performance changes to our data curation. Although stage 1 has already developed strong language modeling capabilities, it precedes the knowledge-intensive tuning of stage 2, providing an ideal balance of baseline capabilities without the risk of catastrophic forgetting of instruction-following abilities during domain adaptation. Notably, the data mix used in stage 1 includes DCLM (Li et al., 2024), which is a dataset obtained by filtering web-data using a classifier trained on instruct-data. Hence, OLMo2-7B already has relatively strong question-answering capabilities after stage 1.

Each Biomed-Enriched variant was trained with exactly 33.6 billion tokens, using identical hyperparameters. We follow the annealing strategy of OLMo2 used in the mid-training phase. By maintaining strict parameter parity across experiments, we created a controlled environment focused solely on measuring the effectiveness of our different data curation strategies.

6.4.2 Encoder Setup

We also conducted continual-pretraining experiments on encoder models. Starting from ModernBERT-base (Warner et al., 2024) after its context extension phase, we followed exactly the same hyperparameters as BioClinical-ModernBERT phase 1. Our baseline reproduces their standard continual pretraining setup on public PubMed/PMC. Our clinical-upsample recipe additionally applies educational filtering (score ≥ 3) and upsamples clinical cases 100× and clinical domain content 10×.

Table 6.4 summarizes the hyperparameters for both setups.

6.4.3 Evaluation

For decoders, our evaluation consists of zero-shot testing on MMLU medical subcategories, MedQA (Jin et al., 2021), MedMCQA (Pal et al., 2022), and PubMedQA (Jin et al., 2019). For the French adaptation assessment, we use 5-shot evaluation on

FrenchMedMCQA (Labrak et al., 2022). We compare our models against three baselines: OLMo2-7B-stage1, Llama-3.1-8B, and Meditron-70B.

For the encoder models, we evaluate on 11 tasks spanning clinical and biomedical domains. Clinical tasks include ChemProt, Phenotype, COS, SocialHistory, and DEID. Biomedical tasks include AnatEM, BC5CDR, JNLPBA, NCBI Disease, GAD, and Hallmarks of Cancer. Following the evaluation recipe of BioClinical-ModernBERT, we finetune for 10 epochs on clinical tasks and 20 epochs on biomedical tasks, with early stopping patience of 3 epochs for all. We compare against BioClinical-ModernBERT (Sounack et al., 2025), which was trained on 169B tokens from PubMed, PMC, and 20 clinical datasets including MIMIC-III, MIMIC-IV, CheXpert Plus radiology reports, and various clinical notes from multiple institutions, most requiring data use agreements.

6.5 Results

6.5.1 Decoder Results

Tables 6.5 and 6.6 present decoder results on medical QA and MMLU benchmarks. The combined strategy BE-All achieves the best average performance (61.08%), with the strongest gains on MedQA (+2.4 pts) and MMLU Clinical Knowledge (+1.5 pts). Different enrichment strategies show complementary strengths: clinical upsampling yields a +4 point improvement on MMLU Professional Medicine, while educational filtering improves Medical Genetics by +2 points and MedMCQA by +1.2 points. BE-All reaches target performance using roughly one-third of the training tokens compared to BE-Base, with stable improvements visible from early checkpoints. BE-French also demonstrates successful adaptation to French medical QA (40.5% vs 38.3% baseline, Figure 6.4), showing that language-specific upsampling generalizes beyond English.

6.5.2 Encoder Results

Tables 6.7 and 6.8 present encoder results across 11 clinical and biomedical benchmarks. Our BE-Clinical recipe achieves 77.08% average F_1, matching BioClinical-ModernBERT (77.05%) while using 2.5× fewer tokens and only public PubMed/PMC data. Adding a decay phase with high-quality educational content brings slight additional gains: the Biomedical decay variant (edu ≥ 4) achieves 77.32%.

Different decay targets show task-specific strengths: Study decay yields the best results on AnatEM (79.7%) and NCBI Disease (81.2%), Review decay excels on GAD (80.2%), and Clinical Case decay achieves the highest SocialHistory score (57.0%). These results suggest that public PMC data has more potential for clinical NLP than commonly exploited, and that targeted curation can close the gap with models trained on restricted datasets.

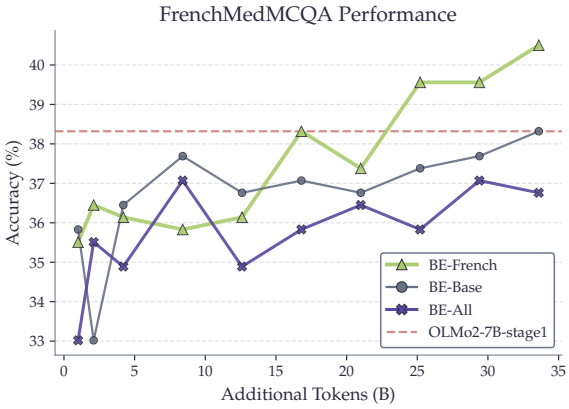

Figure 6.4: **FrenchMedMCQA performance.** BE-French outperforms other variants on French medical QA, demonstrating effective language-specific improvement through targeted upsampling.

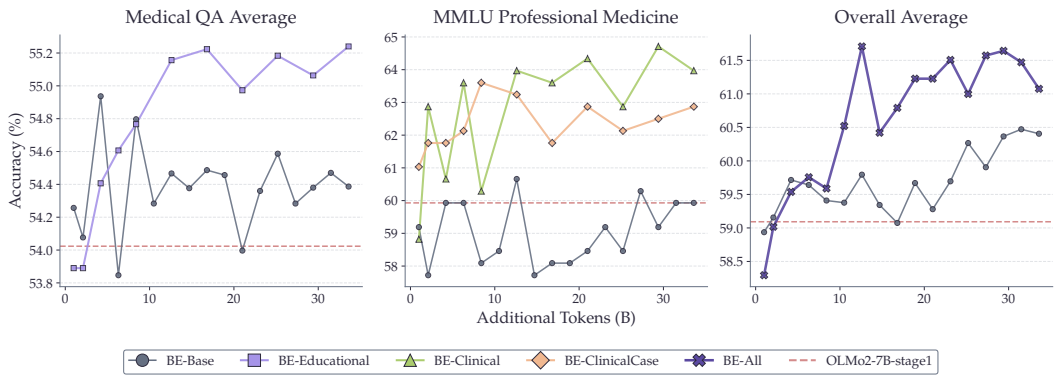

Figure 6.5: **Decoder training progression.** BE-All reaches target performance with approximately one-third of the training tokens required by BE-Base.

6.6 Discussion

The core advantage of paragraph-level annotation lies in its ability to identify valuable content that would be missed by document-level approaches. Clinical case descriptions, for instance, often appear in isolated sections of broader scientific articles. By operating at the paragraph level, we extracted 2 million clinical case paragraphs from PMC Open Access. This content is otherwise difficult to obtain at scale due to privacy restrictions on hospital records.

For decoders, our results reveal clear task-specific benefits from different enrichment strategies. Educational filtering improves knowledge-intensive QA tasks, clinical upsampling enhances clinical reasoning benchmarks, and the combination of both yields the best overall performance with faster convergence. The successful French adaptation through targeted upsampling further demonstrates that this framework

Model	MedQA	MedMCQA	PubMedQA	Avg
Reference Models				
Llama-3-8B	59.70	57.47	74.80	63.99
Meditron-70B	57.10	46.80	76.60	60.17
Our Variants				
OLMo2-7B-stage1	45.33	41.14	75.60	54.02
+ BE-Base	44.85	41.91	76.40	54.39
+ BE-Clinical	41.95	39.35	76.60	52.63
+ BE-ClinicalCase	42.11	39.52	76.60	52.74
+ BE-Educational	45.64	**43.08**	77.00	55.24
+ BE-All (33.6B)	**47.21**	42.79	76.60	55.53
+ BE-All (12.6B)	**47.21**	42.94	76.40	**55.52**

Bold indicates best result per column. Underline: second-best.

Table 6.5: **Decoder results: Medical QA benchmarks.** BE-All achieves the best QA average with both 33.6B and 12.6B tokens.

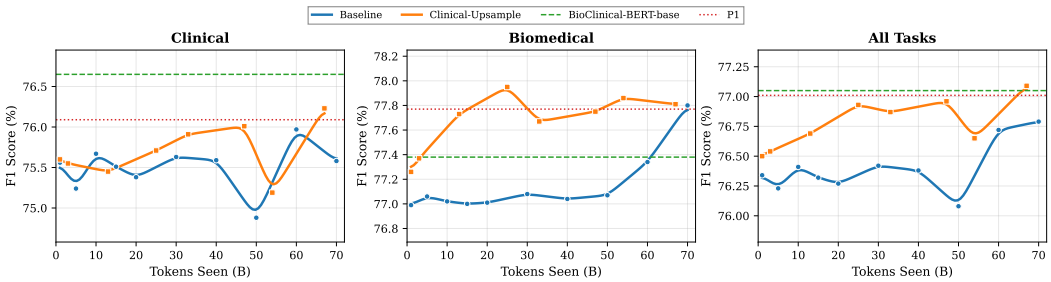

Figure 6.6: **Encoder training curves** on clinical, biomedical, and combined benchmarks. Our clinical-upsample recipe (orange) reaches higher performance faster on clinical tasks. At 67B tokens, we match BioClinical-ModernBERT (green dashed) which required 169B tokens and 20 clinical datasets.

generalizes beyond English. However, aggressive clinical upsampling can reduce performance on general biomedical tasks like College Biology, suggesting a trade-off between specialization and breadth.

For encoders, the decay phase experiments reveal that different paragraph types benefit different downstream tasks: Study paragraphs improve NER performance, Clinical Case paragraphs improve social history extraction, and Review paragraphs excel on relation extraction. The distinct task-specific strengths of each decay variant suggest a promising direction: model merging or checkpoint averaging could combine the benefits of specialized training runs without requiring additional compute. Rather than training general-purpose models on massive undifferentiated corpora, these findings support a more modular approach where data composition is tailored to downstream requirements.

Model	Anat	Clin	Bio	Med	Gen	Prof	Avg
Reference Models							
Llama-3-8B	68.89	74.72	78.47	61.85	83.00	70.22	72.86
Meditron-70B	53.30	66.70	76.30	63.00	69.00	71.60	66.65
Our Variants							
OLMo2-7B-stage1	54.81	63.40	69.44	53.18	69.00	59.93	61.63
+ BE-Base	57.04	64.15	**70.83**	**59.54**	69.00	59.93	63.42
+ BE-Clinical	53.33	63.40	65.28	58.38	66.00	**63.97**	61.73
+ BE-ClinicalCase	57.04	64.91	66.67	**59.54**	69.00	62.87	63.34
+ BE-Educational	57.04	65.28	68.06	56.65	**71.00**	58.82	62.81
+ BE-All (33.6B)	60.00	**65.66**	68.06	58.96	69.00	61.40	63.85
+ BE-All (12.6B)	**64.44**	64.53	68.75	57.23	**71.00**	62.87	**64.80**

Anat=Anatomy, Clin=Clinical Knowledge, Bio=College Biology, Med=College Medicine, Gen=Medical Genetics, Prof=Professional Medicine.

Table 6.6: **Decoder results: MMLU Medical subcategories.** BE-All (12.6B) achieves the best MMLU average despite using one-third of the tokens.

6.7 The Risks of Neural Quality Filtering

[1]

The results above show that selecting content by type and educational quality can improve pretraining efficiency. A natural next step is to apply neural quality classifiers, as done by FineWeb-Edu (Penedo et al., 2024) and DCLM (Li et al., 2024), to filter pretraining data more aggressively. However, this raises a question: do these classifiers treat all content equally, or do they have blind spots?

To investigate this, we designed Benchmark-In-A-HayStack (BiaHS), an experiment to test how different text-quality classifiers rank benchmark-like content within a large web corpus. We sampled 35 benchmark instances from three datasets: 15 samples from MMLU (3 samples each from anatomy, computer security, high school geography, moral scenarios, and college physics), 10 from GSM8K (Cobbe et al., 2021), and 10 from GPQA (Rein et al., 2023). Each benchmark sample was formatted as a standalone document containing the question, answer choices, and reference solution. These 35 benchmark documents were inserted as independent entries into a corpus of 99,965 documents sampled from FineWeb, yielding a final evaluation set of 100,000 documents where benchmarks constituted 0.035% of the total.

We scored all documents using four quality classifiers: DCLM (FastText-based, used to filter OLMo 2 training data), Textbook (FastText-based, used to filter OLMo 1 training data), FineWeb-Edu (transformer-based), and the GAPeron classifier (transformer-based). Each classifier produced a full ranking of all documents based on their predicted quality score.

[1]This section is based on our contribution to Godey et al. (2026).

Model	ChemPr	Pheno	COS	Social	DEID	Avg
Baselines						
ModernBERT-base	89.5	48.4	94.0	53.1	78.3	72.66
BioClinical-MdnBERT†	90.0	**60.7**	94.8	56.0	**81.8**	**76.66**
Our Models (Public Data Only)						
Baseline	89.8	59.6	94.5	55.5	78.5	75.58
BE-Clinical	89.9	59.9	94.8	56.8	79.7	76.22
With Decay Phase						
+ Review	**90.3**	59.7	94.6	55.4	79.6	75.92
+ Study	**90.3**	**60.3**	94.7	55.6	78.6	75.90
+ Clinical Case	89.5	59.4	94.5	**57.0**	79.3	75.94
+ Biomedical	90.1	59.9	**94.9**	56.4	79.6	76.18

† Trained on 169B tokens including 20 clinical datasets (MIMIC-III/IV, CheXpert, etc.). Our models use only public PubMed/PMC.
ChemPr=ChemProt, Pheno=Phenotype, Social=SocialHistory.

Table 6.7: **Encoder results: Clinical tasks.** Our BE-Clinical recipe matches BioClinical-ModernBERT on clinical tasks using only public data.

The results reveal a striking pattern. The DCLM classifier ranks all MMLU and GSM8K samples in the top-5 percentiles. If only the top 5% of documents are selected through filtering, the probability of encountering a benchmark sample in a training batch increases by a factor of approximately 20. The FineWeb-Edu classifier also ranks benchmark samples consistently higher, likely because its annotation prompt explicitly asks for documents with high educational value, which naturally favors exam-style items and step-by-step solutions.

The DCLM classifier pushes benchmark samples even higher and more consistently. Its FastText model is trained to separate synthetically generated instructions from high-scoring Reddit posts, from general web text. This training objective naturally favors solved Q&A structures with a short question, a direct answer, and brief reasoning, which aligns with the format of common benchmark datasets.

In contrast, the GAPeron classifier does not significantly push benchmark samples. It was trained on annotations focused on general content quality across multiple dimensions: accuracy, clarity, coherence, grammar, depth of information, and overall usefulness. This broader framing does not specifically reward the structured question-answer format typical of benchmark samples.

Implications. This experiment suggests that the way quality classifiers are trained can strongly influence contamination risks. As neural quality filtering becomes a standard step in data curation, these design choices deserve closer examination. For the Biomed-Enriched work presented earlier in this chapter, we rely on content-type

Model	AnatEM	BC5CDR	JNLPBA	NCBI	GAD	HoC	Avg
Baselines							
ModernBERT-base	77.2	87.9	74.3	77.7	76.8	66.6	76.75
BioClinical-MdnBERT†	**79.2**	88.7	74.8	78.7	75.8	67.0	77.37
Our Models (Public Data Only)							
Baseline	78.8	88.7	74.8	**81.3**	76.2	67.0	77.80
BE-Clinical	78.5	88.1	74.6	79.4	77.7	68.5	77.80
With Decay Phase							
+ Review	78.5	88.5	74.7	79.6	**80.2**	68.8	78.38
+ Study	**79.7**	88.4	**74.8**	81.2	78.7	67.7	**78.42**
+ Clinical Case	78.7	88.3	74.7	80.0	77.9	68.1	77.95
+ Biomedical	79.0	**88.6**	74.7	79.9	78.4	**69.0**	78.27

† Same model as Table 6.7. Trained on 169B tokens with 20 clinical datasets.

Table 6.8: **Encoder results: Biomedical tasks.** Study decay achieves 78.42%, outperforming BioClinical-ModernBERT (77.37%) with only public data.

classification and educational quality scoring rather than general quality filtering, which avoids this particular risk.

6.8 Limitations

Our decoder experiments used 7B parameter models. Larger models may respond differently to data curation choices. The two-stage annotation pipeline relies on XLM-RoBERTa-base, which may have limited capacity for nuanced paragraph classification. Our clinical case paragraphs come from published case reports, which may differ stylistically from actual hospital records. The educational quality scores are neural heuristics rather than gold-standard pedagogical assessments.

For the BiaHS experiment, the evaluation was conducted on a relatively small number of benchmark samples (35) inserted into a single web corpus (FineWeb). While the results are directionally clear, a larger-scale study across multiple corpora would strengthen the conclusions.

Conclusion

Two findings emerge from this chapter. First, paragraph-level annotation reveals that scientific articles are far more heterogeneous than their metadata suggests: 91.6% mix content types, and educational quality varies by 2.6 points within a single article. Selecting the right paragraphs allows matching full-corpus performance with a third of the tokens, and extracting 2 million clinical case paragraphs provides public clinical text without data use agreements.

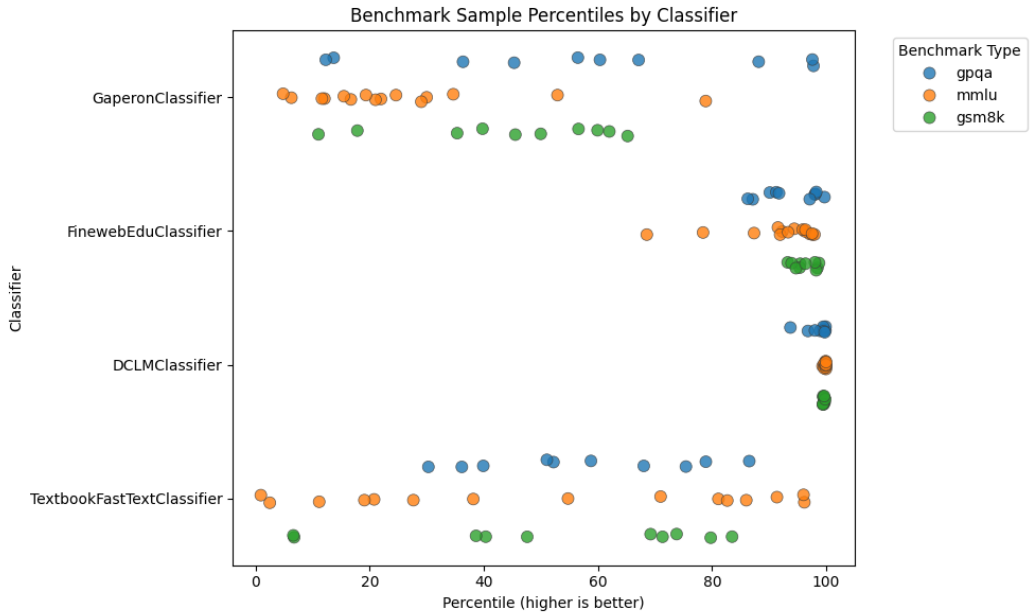

Figure 6.7: **BiaHS: Benchmark-In-A-HayStack results.** Percentile rank of 35 benchmark samples (MMLU, GSM8K, GPQA) within 100K FineWeb documents, as scored by four quality classifiers. DCLM ranks all benchmark samples in the top-5 percentiles.

Second, not all filtering is created equal. Neural quality classifiers trained on instructional or educational content can amplify benchmark contamination by a factor of 20, effectively turning a data curation step into a memorization accelerator. Content-type classification, by contrast, selects for relevance rather than format, and does not carry this risk.

We now have both a general biomedical corpus (*biomed-fr*, Chapter 5) and a method for selecting the most relevant paragraphs within it. The next chapter addresses how to combine these resources into a curated training mix, and how quality signals can be used at scale to build a better corpus for pretraining ModernCamemBERT-bio.

7 Which Quality Signals Matter?

7.1 Introduction

Here is one passage as it came out of an ISTEX PDF:

```
… vise ~ remonter l&apos;organe prolab6 (correction de la pt6se) et
h le soutenir dans sa position id6ale plus qu&apos;/~ le fixer (fixation).
Ce sout~nement est assur6 par le renforcement du p6rin6e grace aux
tissus naturels que l&apos;on r6pare (rapphie) ou que l&apos;on consolide
avec du mat6riel local (plastie, ligamentopexie)…
```

And here is what it should read:

> … la chirurgie vise à remonter l'organe prolapsé (correction de la ptose) et à le soutenir dans sa position idéale plus qu'à le fixer (fixation). Ce soutien est assuré par le renforcement du périnée grâce aux tissus naturels que l'on répare (raphie) ou que l'on consolide avec du matériel local (plastie, ligamentopexie)…

OCR had read every *é* as a *6*, apostrophes came as `l'`, and tildes had drifted into the middle of words. Multiply this by the hundreds of thousands of similar PDFs in HAL and ISTEX, and assembling a 10-billion-token biomedical training mixture stops being a collection problem and becomes a repair problem.

Chapter 5 introduced *biomed-fr*, a 413-million-word public French biomedical corpus. Chapter 6 showed that paragraph-level annotation by a large language model can identify content types and educational quality more precisely than document-level filters, and that neural quality classifiers carry contamination risks. This chapter applies these lessons at scale. We assemble *biomed-fr-v2*, a 10-billion-token mixture from six French sources, and ask which quality signals actually improve downstream performance once the dust settles. The corpus is the training data for ModernCamemBERT-bio, presented in Chapter 9.

The contributions of this chapter are: (1) a multi-source French biomedical corpus 25 times larger than *biomed-fr*, including a curation pipeline that handles the OCR artifacts, boilerplate, and structural noise of HAL and ISTEX; (2) an LLM-based annotation of 2.16 million paragraphs along four quality signals and nine content types; (3) ablation studies that quantify the effect of each signal and each type on downstream performance, with one notable surprise: writing-quality and terminological-precision filters *degrade* performance.

7.2 Sources

We assemble *biomed-fr-v2* from six complementary French-language sources. HAL, ISTEX, and FineWiki-bio cover scientific biomedical literature. EMEA contributes drug package inserts. E3C contributes clinical case reports. Synthetic textbooks generated from medical ontologies complete the mixture with the vocabulary of French administrative nomenclatures. The final composition is summarized in Table 7.4.

- **HAL.** Medical theses, research articles, and reports deposited on the HAL open archive, filtered to the biomedicine section.
- **ISTEX.** French scientific articles from the ISTEX corpus, filtered to biology and medicine.
- **FineWiki-bio.** Biomedical subset of FineWiki, the Wikipedia counterpart to FineWeb (Penedo et al., 2024), restricted to French and filtered for the biomedical domain.
- **EMEA** (Névéol et al., 2014). Drug package inserts published in French by the European Medicines Agency.
- **E3C** (Magnini et al., 2020). Multilingual clinical corpus (layer 3, French partition), composed of clinical cases extracted from medical journals and theses.
- **Synthetic ontology textbooks.** Educational text generated from three French medical ontologies published by the Agence du Numérique en Santé in RDF/OWL format: CIM-10 (diagnoses, 19,161 codes), CCAM (procedures, 38,191 codes), and ATC (medications, 6,950 codes). Random walks through the ontology graph (1.3 million walks total) traverse hierarchical relations and, for CIM-10, causal and exclusion relations. A large language model (Qwen3-235B-A22B-Instruct) reformulates each walk into a paragraph of medical prose while preserving the original codes and relations. The full pipeline is described in Chapter 10.

HAL and ISTEX are distributed in heterogeneous formats: some documents come with structured XML, others only as PDFs, sometimes scans of printed material. For documents without clean XML, we apply the extraction and cleaning pipeline described in Section 7.3. FineWiki-bio, EMEA, E3C, and the synthetic textbooks are already of high quality and are integrated directly into the final mixture, bypassing this pipeline.

7.3 Curation Pipeline

A portion of HAL and ISTEX documents come with clean XML and proceed directly to annotation. The other portion is available only as PDFs, sometimes scans of printed material (particularly for articles published before the 1990s). Once converted to text, these PDFs contain OCR artifacts (incorrect ligatures, character substitutions, fragmented words), bibliographic references, repeated headers, and irrelevant content. We therefore apply an LLM-based extraction and cleaning pipeline, described below. The other sources (FineWiki-bio, EMEA, E3C, synthetic textbooks) are already clean and bypass this pipeline entirely.

Extraction and cleaning. Each document is split into segments of approximately 5,000 tokens at paragraph boundaries. A large language model (Qwen3-Next-80B-A3B-Instruct) receives each segment and produces structured JSON output containing semantically complete passages, with references, metadata, headers, and captions removed, and OCR errors corrected (ligatures, encoding). Extracted passages must contain between 30 and 600 words and at least 60% alphabetic characters. We then apply Jaccard-similarity deduplication (threshold 0.80, sliding window of 5 passages), followed by the FineWeb-2 filter chain for French: encoding correction, repetition filters, and Gopher and FineWeb quality filters.

Fidelity verification. To ensure the LLM does not introduce content absent from the source, we measure character-level overlap between each extracted passage and its source document on 300 ISTEX passages (after typographic normalization). 94.4% of passages have $\geq$ 90% overlap and 84.1% have $\geq$ 99% overlap; the median rate of words absent from the source is zero (mean: 3.4%). Manual inspection of edge cases shows that divergences come from OCR corrections (such as the surgical passage shown at the start of this chapter) or from converting tables into prose. The opposite case is also common: a clean 1,143-character passage on cancer-related depression is extracted with 100% character overlap and no additional words introduced. These two cases bracket the typical behavior of the pipeline: it cleans aggressively when the source is corrupted by OCR, and copies verbatim when the source is already clean.

LLM quality annotation. The same LLM annotates each passage along four quality signals scored from 1 to 10: educational value (`edu`, mean 6.5), content richness (`cont`, 6.8), writing quality (`writ`, 7.6), and terminological precision (`term`, 7.6). The codes in parentheses are the abbreviations used in ablation tables. A content type is also assigned among nine categories: `research_findings` (526k passages), `medical_knowledge` (388k), `clinical_guidance` (218k), `research_methodology`,

Type removed	Avg F_1	Δ
Removal helps		
`drug_information`	64.35	+0.67pp
`research_findings`	64.24	+0.56pp
`policy_administrative`	64.07	+0.39pp
Controls		
Random (same volume)	63.72	+0.04pp
Full corpus (reference)	63.68	ref.
Removal hurts		
`medical_knowledge`	63.55	−0.13pp
`research_methodology`	63.51	−0.17pp

Table 7.1: **Content-type ablation.** We remove one type and measure Δ relative to the full corpus.

`drug_information`, `background_review`, `policy_administrative`, `patient_case` (36k), and `other`. In total, 2.16 million passages are annotated.

7.4 Content-Type Ablation

We measure the effect of each content type by ablation: we remove one type and compare against the full corpus on three evaluation tasks (DiaMed, FrACCO-30, and EMEA, 9 seeds). Table 7.1 shows that removing `drug_information` *improves* performance by +0.67pp, while removing `medical_knowledge` degrades it by −0.13pp. A random control (removing the same amount of text at random) gives +0.04pp, confirming that the effect is specific to the content type.

We exclude `drug_information` from the final corpus (redundant with the EMEA source already included separately) and `other`. Paragraphs shorter than 50 tokens are also excluded.

7.5 Quality Filtering and Upsampling

We then test the effect of thresholding on the quality signals (Table 7.2). Joint filtering with `edu`$\geq$ 7 and `cont`$\geq$ 7 produces a +0.48pp gain; a stricter threshold (`edu`$\geq$ 8 and `cont`$\geq$ 8) reaches +0.55pp. By contrast, the writing-quality and terminological-precision signals *degrade* performance when used as filters (−0.26 and −0.33pp). This degradation likely comes from the fact that these filters eliminate informal but informative clinical documents, such as case notes and lightly edited reports.

Rather than applying a strict filter, we adopt a quality-weighted upsampling inspired by FineWeb (Penedo et al., 2024). Each article receives a quality ratio, defined as the proportion of its paragraphs satisfying `edu`$\geq$ 7 AND `cont`$\geq$ 7. Articles are

Filter	Avg F_1	Δ
Helpful filters		
`edu`≥ 8 AND `cont`≥ 8	**59.37**	+0.55pp
`edu`≥ 7 AND `cont`≥ 7	59.30	**+0.48pp**
`edu`≥ 6 AND `cont`≥ 6	58.95	+0.13pp
No filter (reference)	58.82	ref.
Harmful filters		
`term`≥ 7	58.57	−0.26pp
`writ`≥ 7	58.49	−0.33pp

Table 7.2: **Quality-signal ablation.** Educational value and content richness improve performance; writing quality and terminological precision degrade it.

Ratio of `edu`≥ 7 AND `cont`≥ 7 paragraphs	Coefficient
0%	excluded
1–25%	4.3×
26–50%	8.5×
51–75%	12.8×
76–99%	21.3×
100%	34.1×

Table 7.3: **Upsampling coefficients** per article-quality bucket.

then upsampled in proportion to this ratio (Table 7.3). Articles with 0% high-quality paragraphs are excluded from the final corpus.

7.6 Final Composition

The final training mixture, after cleaning and quality-weighted upsampling, contains 10 billion tokens (Table 7.4). These 10 billion include the repetitions induced by upsampling (coefficients ranging from 4.3× to 34.1× depending on quality, see Section 7.5); the underlying unique content is more limited. HAL and ISTEX articles (passed through the pipeline of Section 7.3) and FineWiki-bio (integrated as-is) are grouped under "curated biomedical articles." The three other sources (EMEA, E3C, synthetic textbooks) are also integrated as-is.

Conclusion

Two findings emerge from the ablations. First, not all content types contribute equally: removing drug information actually improves downstream performance, while removing medical knowledge degrades it. Second, not all quality signals

Source	Tokens	%	Content
Curated biomedical articles	7B	70%	HAL + ISTEX + FineWiki-bio
Synthetic ontology textbooks	2B	20%	CIM-10, CCAM, ATC
EMEA (drug inserts)	600M	6%	Pharmacology
E3C (clinical sentences)	400M	4%	Patient cases
Total	**10B**	**100%**	

Table 7.4: **Composition of the *biomed-fr-v2* training mixture** after quality-weighted upsampling.

work as filters: educational value and content richness help, but writing-quality and terminological-precision filters *hurt* performance, likely because they remove the informal but informative clinical documents that ultimately matter most. The lesson, perhaps, is that curation should be guided by what is being said, not by how well it is written.

The corpus is now ready. The next two chapters use it to train French biomedical encoders. Chapter 8 starts with the simplest recipe (continual pretraining with MLM on *biomed-fr*) and shows that even a modest public corpus produces a competitive model. Chapter 9 returns to *biomed-fr-v2* with a more ambitious recipe: a temporary detour through causal language modeling that, somewhat unexpectedly, leaves a permanent imprint on the encoder.

Part III

Pretraining Language Models

8 Encoder Models for French Biomedicine

8.1 Introduction

In Chapter 5, we found that 55% of biomedical subwords are absent from CamemBERT's vocabulary. PubMedBERT (Gu et al., 2022a) turned this kind of observation into a principle: for domains with enough unlabeled text, training a language model from scratch with a specialized vocabulary outperforms continual pretraining. The gain over BioBERT (Lee et al., 2020a) was modest, but the conclusion was adopted widely.

For French, Labrak et al. (2023) went further. They reported that continual pretraining from CamemBERT led to F_1 losses of up to 20 points, and released DrBERT, trained from scratch on 128 V100 GPUs for 20 hours. The claim was strong, the compute was expensive, and the implication was clear: continual pretraining does not work for French biomedical models. Yet the size of the reported loss is difficult to reconcile with what we know about continual pretraining. A model that has already seen 138 GB of French text should not lose 20 points from seeing additional biomedical text. Something in the evaluation deserves closer examination.

In this chapter, we show that continual pretraining on *biomed-fr* (Chapter 5) yields an average improvement of +2.54 F_1 across five biomedical NER benchmarks, using 2 V100 GPUs for 39 hours. We also identify the methodological differences that led to the contradictory conclusion.

8.2 Pre-training

For the adaptation of CamemBERT to the biomedical domain, we conducted a second phase of pre-training on both versions of the *biomed-fr* corpus, starting from the weights and configuration of the camembert-base model. We applied the Masked Language Modeling (MLM) task with whole-word masking, following the method of Martin et al. (2020a). We used the Adam optimizer (Kingma and Ba, 2015) with $\beta_1 = 0.9$ and $\beta_2 = 0.98$, and a learning rate of 5×10^{-5}. We performed 50,000 steps over 39 hours using two Tesla V100 GPUs. A batch size of 8 per GPU and gradient accumulation over 16 steps were used to achieve an effective batch size of 256.

8.3 Fine-tuning and Evaluation

Regarding model evaluation, we collected three named entity recognition evaluation datasets. These datasets cover various styles, allowing us to assess the model's versatility across different subdomains of biomedicine.

QUAERO. The QUAERO corpus (Névéol et al., 2014) consists of two evaluation sets: EMEA, containing drug leaflets, and MEDLINE, containing scientific article titles. The entities are manually annotated following 10 semantic groups from the UMLS (Lindberg et al., 1993). As some of these entities are nested, we kept only the entities with the coarsest granularity.

E3C. For evaluation, unlike the *biomed-fr* corpus, we use layers 1 and 2. These layers contain documents of different types, including clinical cases extracted from scientific articles. Layer 2 is semi-annotated, and it is used as the training set for fine-tuning, with 10% dedicated to the validation set. We evaluate on layer 1, which is fully manually annotated. There is only one class, and the objective is to find clinical entities in the text, regardless of their type.

CAS. The CAS corpus (Grouin et al., 2019) also consists of clinical cases from scientific articles. We focus on task 3 of DEFT 2020 (Cardon et al., 2020), which is an information extraction task based on CAS. It includes two subtasks, and thus two sets of annotations. In the first subtask, two classes need to be identified: *pathology* and *signs or symptoms*. The second subtask concerns associated information, including *anatomy*, *dose*, *examination*, *mode*, *timing*, *substance*, *treatment*, and *value*. These two tasks will be referred to as CAS1 and CAS2, respectively.

Fine-tuning. For fine-tuning, we used Optuna (Akiba et al., 2019) for hyperparameter selection. We set the learning rate to 5×10^{-5}, the warmup ratio to 0.224, and the batch size to 16. We performed 2000 steps. Predictions were made using a simple linear layer on top of the model. None of the CamemBERT layers were frozen.

Evaluation. Scores are measured using the seqeval tool (Nakayama, 2018) in strict mode with micro-average and the IOB2 scheme. For each evaluation, the best fine-tuned model on the validation set is selected to measure the final score on the test set. We average the results over 10 evaluations with different seeds.

8.4 Results and Discussion

CamemBERT vs CamemBERT-bio. We observe a significant performance gain on all evaluation datasets with our new model (Table 8.1). On average, we achieve a 2.54-

Dataset	CamemBERT	CamemBERT-bio	
		biomed-fr-small	biomed-fr
Clinical			
CAS1	70.50	72.94	**73.03**
CAS2	79.02	80.00	**81.66**
E3C	67.63	67.96	**69.85**
Leaflets			
EMEA	74.14	75.93	**76.71**
Scientific			
MEDLINE	65.73	65.48	**68.47**
Average	71.40	72.46	**73.94**

F_1 scores (micro-average, seqeval strict, IOB2). Mean over 10 seeds. biomed-fr-s = biomed-fr-small (10% subset).

Table 8.1: **CamemBERT vs CamemBERT-bio** on five biomedical NER benchmarks. Continual pretraining on *biomed-fr* yields +2.54 F_1 on average.

point improvement in F-score. This gain is observed across all styles, demonstrating the model's versatility for both clinical and scientific domains.

BIOMED-FR-SMALL VS BIOMED-FR. We observe a decrease in performance with the biomed-fr-small dataset, but there is still a significant gain on certain datasets compared to CamemBERT. This confirms that the size of the corpus positively influences the performance, even in a specialized domain like biomedicine.

COMPARISON WITH THE STATE OF THE ART. We compared the performance of CamemBERT-bio with various previously mentioned approaches (Table 8.2). CamemBERT-bio achieves the best results for almost all evaluation datasets. Dura et al. (2022) did not observe improvement on EMEA and MEDLINE compared to CamemBERT because their pre-training corpus consists of documents from APHP, making it a less diverse corpus. However, they gain several points on their evaluation dataset, which is also based on APHP documents. van Mulligen et al. (2016) presents the highest score on MEDLINE and the best recall on EMEA. Their approach is based on a knowledge-based model, allowing them to achieve the best recall on both QUAERO evaluation datasets. Furthermore, their approach is the only one in this table capable of handling nested entities, giving them an advantage.

It is important to note that these different CamemBERT-based approaches have various experimental setups. The presence of CRF layers instead of a simple linear layer after CamemBERT, freezing CamemBERT layers, and variations in hyperpa-

Authors	CAS1	CAS2	EMEA	MEDLINE
seqeval				
Dura et al. (2022)	–	–	72.90	59.70
Ours	**73.03**	**81.66**	**76.71**	**68.47**
BRATeval				
Le Clercq de Lannoy et al. (2022)	–	–	67.40	55.30
van Mulligen et al. (2016)	–	–	74.90	**69.80**
Copara et al. (2020)	61.53	73.70	–	–
Ours	**84.97**	**83.25**	**77.80**	56.16

F_1 scores. BRATeval is the evaluation tool from the CLEF eHealth 2016 campaign (Névéol et al., 2016).

Table 8.2: **Comparison with existing approaches** on four NER tasks, using two evaluation tools.

rameters are examples of differences observed in addition to the pre-training corpus, which makes the comparison more challenging.

8.5 Evaluating Models: Methodology Matters

In a recent work, Labrak et al. (2023) introduced a new French biomedical language model named DrBERT. The authors contend that performing continual-pretraining on biomedical data from CamemBERT leads to reduced performance. They used a different methodology for the evaluation of named entity recognition, prompting us to examine the implications of each approach on the interpretation of the results.

Our evaluation approach is centered around micro F_1, which is measured using seqeval in strict mode with the IOB2 scheme. Their approach is based on weighted F_1 with independent token classification. Every token has a label and the model is evaluated for each of them, whereas with seqeval, the model is evaluated for each entity. Notably, the "O" token, representing non-entity, is the most frequent token and thus holds significant weight in the evaluation process. As all tokens are labeled, the number of predictions depends on the tokenizer, thereby influencing the final score.

In order to explore the impact of token labeling variations on the evaluation, we conducted an additional experiment on EMEA and MEDLINE where we reproduced the DrBERT methodology that we named *token-with-O*, and another one where we excluded the "O" token and only labeled the first token of each entity, referred to as *entity-without-O*.

We observe significant differences in scores between the two methodologies (Table 8.3). The methodology *token-with-O* consistently reports higher scores across all tasks compared to the alternative method, *entity-without-O*. Notably, the best-

Method	Model	EMEA		MEDLINE	
		w-F_1	m-F_1	w-F_1	m-F_1
token-with-O					
	DrBERT-7GB	87.45	34.95	75.52	**15.07**
	CamemBERT-bio	**90.37**	36.27	**77.89**	14.82
	CamemBERT	88.33	**47.45**	76.20	11.92
entity-without-O					
	DrBERT-7GB	66.72	**24.72**	60.70	**10.80**
	CamemBERT-bio	**73.53**	24.15	**62.04**	8.70
	CamemBERT	71.85	22.71	60.95	9.41

w-F_1 = weighted F_1, m-F_1 = macro F_1. Averaged over 10 runs.

Table 8.3: **Impact of evaluation methodology.** The best model varies depending on the metric and the treatment of the "O" token.

Model	Time	Hardware	GPU-h	kg CO_2
DrBERT	20h	128× V100	2,560	26.11
AliBERT	20h	48× A100	960	8.16
CamemBERT-bio	39h	2× V100	78	**0.80**

Table 8.4: **Carbon emissions during training.** CamemBERT-bio emits 32× less CO_2 than DrBERT. Estimation based on 34g CO_2eq. per kWh (France, 2022–2023).

performing model varies depending on the chosen methodology for one metric. CamemBERT achieves the highest macro F_1 score on the EMEA dataset, whereas with *entity-without-O*, DrBERT-7GB emerges as the top model. This observation underscores the influential role of the "O" class in determining the best-performing model.

In terms of macro F_1, DrBERT consistently outperforms CamemBERT-bio when evaluated using the *entity-without-O* methodology. On the other hand, CamemBERT-bio exhibits better performance across all other metrics. This indicates that DrBERT demonstrates a more balanced performance across different classes.

Furthermore, Labrak et al. (2023) suggested that continual-pretraining from a French model on French biomedical data is not effective, as it results in an F_1-score loss of up to 20 points. However, considering our success with continual-pretraining and the points we raised about the methodology, we suggest a reassessment of these findings.

8.6 Environmental Impact

Considering estimated carbon emissions during training, AliBERT emits 10 times the amount of CamemBERT-bio, while DrBERT releases 32 times more (Table 8.4). While a direct comparison with AliBERT was not possible, the minor performance variance with DrBERT, set against the pronounced disparities in computational and environmental costs, leads us to advocate for continual-pretraining as the preferred adaptation method for biomedical language models.

8.7 Limitations

Our evaluation focused on one task, named entity recognition. This might mean our understanding of how well our model performs is restricted. Future studies should aim to assess our model across a wider range of tasks and datasets to get a clearer picture of its strengths and weaknesses.

Additionally, the corpus limitations discussed in Chapter 5 apply here as well: *biomed-fr* is dominated by scientific literature and does not contain genuine clinical notes. Models pretrained on this corpus may still underperform on real hospital data compared to models trained on private clinical corpora (Dura et al., 2022).

Conclusion

The vocabulary mismatch identified in Chapter 5 is real, but it does not prevent effective adaptation: +2.54 F_1 at 32× less CO_2 than the from-scratch alternative. The 20-point loss reported by Labrak et al. (2023) turns out to be an artifact of the evaluation protocol, not a property of continual pretraining itself.

Continual pretraining works for encoders. The bidirectional nature of MLM may make these models particularly robust to domain shifts. But the field has moved toward autoregressive decoders, where the dynamics of continual pretraining are different, and the results, as the next chapter discusses, are less encouraging.

9 Beyond Masked Language Modeling

9.1 Introduction

Masked language modeling sees only 15% of its input. At every training step, 85% of the tokens receive no gradient signal. Causal language modeling, by contrast, predicts every token from its predecessors: every position contributes, every position learns. The price is that it can only look backward. The question is whether an encoder can benefit from this dense training signal without permanently losing its ability to look in both directions.

In this chapter, we show that it can. We introduce a two-phase pretraining strategy for ModernCamemBERT-bio: first, continual pretraining with a causal language modeling objective, then a short "decay" phase that converts the model back to masked language modeling. The resulting encoder outperforms the standard MLM-only approach by +2.9 F_1 on average across 8 French biomedical tasks (Base) and +1.5 F_1 (Large), winning on all 8. The effect transfers to English at three scales, with gains of +0.3 to +0.8 F_1 depending on model size.

The CLM phase leaves a lasting imprint on low transformer layers. CKA analysis shows that layers 0–7 diverge 5–44× more between CLM and MLM models than seed noise alone, and this divergence persists through extended MLM decay. Freeze interventions establish a double dissociation: freezing low layers during CLM eliminates the downstream benefit; freezing mid layers preserves it. The advantage cannot be localized post hoc: transplanting layer groups from CLM to MLM recovers at most 35% of the gap. The effect is emergent, requiring the coordinated interaction between all layers.

9.2 Motivation

9.2.1 When Decoder Continue-Pretraining Stops Working

9.2.2 Architecture and Objective

CamemBERT-bio (Chapter 8) demonstrated that continual pretraining works for biomedical encoders. However, it used the original CamemBERT architecture: 512-token context, absolute position embeddings, no Flash Attention. Clinical documents are often longer than 512 tokens, and the architecture has not kept pace with recent advances.

Figure 9.1: **The CLM detour pipeline.** Starting from a pretrained MLM encoder, we continue pretraining with a causal objective (Phase 1), then decay back to MLM (Phase 2). The model does not return to an MLM-like state.

ModernBERT (Warner et al., 2024) addresses these limitations with Flash Attention, RoPE position embeddings, alternating local/global attention, and an 8,192-token context window. For biomedical adaptation, we start from ModernCamemBERT (French) and ModernBERT (English), in both Base (150M) and Large (350M) sizes.

The standard approach would be to continue MLM pretraining on biomedical data, as we did for CamemBERT-bio. But MLM has a structural limitation: only 15% of tokens are masked and receive gradient signal at each step. Causal language modeling does not have this limitation: every token predicts the next, and every position contributes to the loss. The idea is to use CLM for the compute-intensive pretraining phase, and switch back to MLM only at the end, to recover bidirectional attention. The key question is whether this switch erases the benefits of CLM or preserves them.

Related approaches. A handful of recent works explore objective switching for encoder pretraining. Gisserot-Boukhlef et al. (2025) pretrain encoders *from scratch* (210M–1B parameters, 100B tokens) and find that a biphasic CLM-then-MLM schedule outperforms pure MLM under fixed compute. Ettin et al. (2025) train matched encoder/decoder pairs (up to 1B parameters, 1.7T tokens) and show that continued pretraining on the reverse objective does not bridge the encoder-decoder performance gap. AntLM (Team, 2024) alternates between CLM and MLM epochs at small scale (10M words). Our approach differs in three ways: we study the round-trip MLM→CLM→MLM with continued pretraining (not from scratch); we focus on the lasting effect of the CLM phase after returning to MLM; and we provide causal evidence via freeze interventions of where the benefit comes from. As Gisserot-Boukhlef et al. (2025) note, understanding how training objectives transfer to downstream tasks "remains largely unexplored."

9.3 Method

9.3.1 The CLM Detour

Our approach consists of two phases, illustrated in Figure 9.1.

Phase 1: CLM pretraining. Starting from a pretrained ModernBERT encoder, we replace the bidirectional attention mask with a causal mask and train with next-token prediction. The model architecture is unchanged: only the attention mask and loss computation differ. This forces the model to compress information into each token representation, since it cannot look ahead.

Phase 2: MLM decay. After CLM pretraining, we restore bidirectional attention and train with MLM at 15% masking for approximately 10% of the Phase 1 budget. The optimizer state is kept between phases; only the learning rate scheduler resets. Phase 2 decays the learning rate from peak to 10% of peak.

Controlled comparison. The MLM baseline follows the same two-phase structure with 30% masking in Phase 1 and 15% in Phase 2, identical schedule and optimizer. Both trajectories use identical architecture, data, and total compute:

$$\theta_A = T_{\text{MLM}}(T_{\text{CLM}}(\theta_0, n), 0.1n) \qquad \text{(CLM detour)}$$
$$\theta_B = T_{\text{MLM}}(\theta_0, 1.1n) \qquad \text{(MLM baseline)}$$

The only difference is the Phase 1 objective. Any performance gap is attributable to the training objective alone.

9.3.2 Freeze Interventions

To test which layers carry the CLM benefit, we run three freeze experiments on the 22-layer French Base model, where the CLM-MLM gap is largest (+2.9pp):

- **Experiment 1** (low freeze, CLM phase): Layers 0–7 frozen during CLM, then normal decay. Tests whether low-layer modification during CLM is necessary.
- **Experiment 2** (low freeze, decay phase): Normal CLM, then layers 0–7 frozen during decay. Tests whether low-layer CLM changes persist without further updates.
- **Experiment 3** (mid freeze, CLM phase): Layers 8–14 frozen during CLM, then normal decay. Together with Experiment 1, tests selectivity: if freezing low layers eliminates the benefit while freezing mid layers preserves it, this constitutes a double dissociation.

9.3.3 Data

French. We compile 10B tokens (Base) or 25B tokens (Large) from four sources. The main source (7B tokens) is French biomedical literature (scientific articles, clinical guidelines, medical theses), where each paragraph is scored for educational

value using Qwen3-235B and articles are upsampled based on their proportion of high-scoring paragraphs, following the Biomed-Enriched approach (Chapter 6). The remaining sources are synthetic medical QA from French coding systems (2B), clinical cases from the European Clinical Case Corpus (E3C, 400M), and drug package inserts from the European Medicines Agency (600M).

English. For the 50B scale, we mix three components: 60% biomedical literature from Biomed-Enriched (PMC Open Access articles filtered by educational value $\geq$ 3.0), 20% medical instruction-following datasets (MedS-Ins, ReasonMed, MediFlow, UltraMedical, EHR-Ins-Reasoning), and 20% MIMIC-III discharge summaries and clinical notes. The smaller 10B variant uses Biomed-Enriched with clinical case paragraphs upsampled 100× and other clinical documents 10× (80%), plus MediFlow medical instructions (20%), without MIMIC.

9.3.4 Training Details

All runs use decoupled AdamW with peak lr 2×10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.98$, weight decay 10^{-5}, and a global batch size of 384 sequences (~3.1M tokens). Phase 1 uses linear warmup over 100M tokens then constant learning rate. Documents are packed into 8,192-token sequences with end-of-sequence tokens between documents. Training uses bf16 mixed precision on 4×H100 GPUs.

9.3.5 Fine-tuning Protocol

French tasks. Classification (DiaMED) and multilabel tasks (MedDialog, FrACCO, CANTEMIST, DisTEMIST) use a linear head on the CLS token, trained for 15 epochs with lr $= 2 \times 10^{-5}$, batch size 4, and 10% warmup. Multilabel tasks use max sequence length 4096 with weighted BCEWithLogitsLoss; classification uses 2048. NER tasks (EMEA, Medline) use a span-based architecture with BiLSTM, CRF, and biaffine scorer, trained for 4000 steps with lr $= 5 \times 10^{-5}$ for the encoder and 10^{-3} for the head, batch size 16, max length 1024. All tasks use AdamW with weight decay 0.01 and select the best checkpoint by validation F_1.

English tasks. We follow the BioClinical-ModernBERT evaluation protocol (Sounack et al., 2025): lr $= 5 \times 10^{-5}$, weight decay 0.01, batch size 16, 10 epochs for most tasks (20 for NER). All models are fine-tuned with the same hyperparameters per task.

9.3.6 Evaluation

We evaluate on 8 French and 11 English biomedical tasks (Table 9.1), using 9 seeds for French and 5 for English. French baselines include ModernCamemBERT, DrBERT (Labrak et al., 2023), CamemBERT-bio, and CamemBERT (Martin et al.,

Table 9.1: **Evaluation tasks.** Max sequence length reflects the fine-tuning configuration; longer values indicate tasks with longer input documents.

Task	Type	Lang	Max len.
French			
DiaMED	Classification	FR	2048
MedDialog-FR	Multilabel	FR	4096
FrACCO (30/100)	Multilabel	FR	4096
CANTEMIST	Multilabel	FR	4096
DisTEMIST	Multilabel	FR	4096
QUAERO (EMEA/Medline)	NER	FR	1024
English			
ChemProt	Relation Extr.	EN	512
Phenotyping	Classification	EN	8192
COS	NER	EN	512
Social History	NER	EN	512
De-identification	NER	EN	8192
AnatEM	NER	EN	512
BC5CDR	NER	EN	512
JNLPBA	NER	EN	512
NCBI Disease	NER	EN	512
GAD	Classification	EN	512
HoC	Classification	EN	512

2020a). English baselines include ModernBERT, PubMedBERT (Gu et al., 2022a), BioBERT (Lee et al., 2020a), and BioClinical-ModernBERT (Sounack et al., 2025).

9.4 Results

9.4.1 French Biomedical Tasks

Table 9.2 presents the French results. For Base, the CLM detour achieves 60.8% average F_1, outperforming the MLM baseline on all 8 tasks (+2.9pp). For Large, CLM reaches 63.3% versus 61.8% for MLM (+1.5pp). CamemBERT-bio and CamemBERT average 36% and 35% overall, limited by their 512-token context on long clinical documents. We release the CLM-detour models as ModernCamemBERT-bio (Base and Large).

9.4.2 English Biomedical Tasks

Table 9.3 shows the English results at three scales. CLM outperforms MLM on average at all scales, with the gap widening at Large scale (+0.8pp, 7/11 task wins). The smaller English effect is expected: ModernBERT was pretrained on web data that commonly includes biomedical corpora such as PubMed, so its low layers

Table 9.2: **French biomedical results** (macro F_1, 9 seeds). Bold: best per column; underline: second best.

		Multilabel Classification					CLS	NER		
	Ctx	FrACCO-30	FrACCO-100	CANTEMIST	DISTEMIST	MedDialog	DiaMED	EMEA	Medline	Avg
Baselines										
ModernCamemBERT	8192	70.1	55.3	63.3	20.2	60.6	56.4	63.4	55.3	55.6
DrBERT	512	53.0	35.6	37.9	21.4	63.6	57.0	**68.0**	**62.3**	49.9
CamemBERT-bio	512	41.9	20.1	12.8	9.6	38.6	47.7	61.6	56.6	36.1
CamemBERT	512	40.8	19.4	11.9	9.5	37.4	40.6	61.5	54.7	34.5
Our Models (Base, 150M)										
MLM baseline	8192	69.9	56.8	64.9	23.5	62.5	63.4	65.4	56.8	57.9
CLM detour	8192	74.8	60.1	71.0	25.5	63.6	**67.4**	65.9	58.2	60.8
Our Models (Large, 350M)										
MLM baseline	8192	79.4	63.3	72.6	29.1	**64.5**	61.2	66.1	58.0	61.8
CLM detour	8192	**80.7**	**65.4**	**74.4**	**30.4**	**64.5**	64.8	67.0	59.5	**63.3**

Table 9.3: **English biomedical results** across 11 benchmarks (5 seeds). Base at 10B and 50B scales; Large at 50B. Bold: best; underline: second best.

		Clinical Tasks					BigBIO Tasks						
	Ctx	ChemPr	Pheno	COS	Social	DEID	AnatEM	BC5CDR	JNLPBA	NCBI	GAD	HoC	Avg
Baselines													
ModernBERT-base	8192	89.5	48.4	94.0	53.1	78.3	77.2	87.9	74.3	77.7	76.8	66.6	74.9
PubMedBERT	512	90.2	52.0	95.0	48.7	80.4	**83.3**	89.7	74.9	**82.1**	79.3	**71.0**	77.0
BioClinical-MdnBERT	8192	90.0	60.7	94.8	56.0	81.8	79.2	88.7	74.8	78.7	75.8	67.0	77.0
Base 150M (10B tokens)													
MLM baseline	8192	90.4	61.3	94.4	55.5	79.3	78.9	88.3	74.9	79.2	78.6	68.9	77.3
CLM detour	8192	90.3	61.0	94.3	55.7	82.6	79.9	89.1	74.2	80.4	79.3	69.4	77.8
Base 150M (50B tokens)													
MLM baseline	8192	**90.5**	61.6	95.1	55.7	81.6	80.0	88.7	74.9	80.3	78.7	67.1	77.7
CLM detour	8192	90.1	**61.9**	**95.2**	54.2	83.2	81.0	89.1	74.5	80.1	78.8	70.0	78.0
Large 350M (50B tokens)													
MLM baseline	8192	**90.5**	61.0	94.9	55.0	82.3	82.0	89.4	**75.5**	81.8	76.4	67.8	77.9
CLM detour	8192	90.4	61.3	94.7	**56.5**	**84.2**	83.2	**89.8**	75.3	81.7	**79.7**	69.3	**78.7**

already partially encode biomedical features. ModernCamemBERT, by contrast, was pretrained on general French web text without biomedical sources, leaving a larger domain gap and correspondingly stronger CLM benefit (+2.9pp vs +0.3pp). We release the English models as ModernBERT-bio.

9.4.3 Optimal Decay Ratio

We sweep the decay length from 2.5% to 50% of the CLM phase at two scales, evaluating on 3 French tasks (DiaMED, FrACCO-30, FrACCO-100; Table 9.4). At both scales, 10% decay gives the best performance; shorter decay (2.5–4%) scores about 1.0pp below optimal, and longer decay (20–50%) provides no additional benefit. This ratio matches the cooldown lengths used in language model pretraining.

9.4.4 Needle-in-a-Haystack

Long-context encoding is critical in biomedical NLP, where clinical documents routinely span thousands of tokens and key information (diagnoses, coding labels) can

Table 9.4: **Decay ratio sweep** on 3 French tasks (DiaMED, FrACCO-30, FrACCO-100; 9 seeds each). 10% decay is optimal at both scales.

Decay tokens	Decay / Phase 1	CLM+decay F_1	MLM+decay F_1
10B Phase 1 (Base, 150M)			
0.25B	2.5%	71.6%	69.1%
1.0B	**10%**	**72.6%**	70.4%
2.0B	20%	71.9%	70.4%
3.0B	30%	72.1%	69.7%
5.0B	50%	71.7%	70.4%
25B Phase 1 (Large, 350M)			
1.0B	4%	73.2%	70.1%
2.5B	**10%**	**73.5%**	70.5%
5.0B	20%	72.7%	70.1%

appear anywhere in the document. We design a synthetic needle-in-a-haystack task in French to evaluate long-context retrieval without entity-type confounds.

Protocol. A medical fact (the "needle") is inserted into a clinical document (the "haystack") at a controlled position, and the model must determine whether a given query fact is present (binary classification). Negative examples use the same template with different slot values, so the task requires precise retrieval rather than shallow pattern matching. We define 8 French medical fact templates covering drugs, vital signs, lab results, diagnoses, procedures, allergies, treatment duration, and symptoms. Each template has 2–4 slots filled from small lexicons:

- *Le patient a reçu 200 mg de morphine par voie intraveineuse.*
  Distractor: *Le patient a reçu 500 mg de tramadol par voie orale.*
- *La tension artérielle mesurée était de 160/90 mmHg.*
- *Le diagnostic retenu est cirrhose hépatique de stade III.*
- *Le patient rapporte une dyspnée évoluant depuis une semaine.*

We generate 1,500 balanced positive/negative pairs across 5 lengths (512–8,192 tokens) and 3 positions (start, middle, end), split 70/15/15 for train/validation/test. We freeze each encoder and train a 2-layer MLP probe (Dropout → Linear → GELU → Dropout → Linear) on CLS representations for 3 epochs (lr = 2×10^{-5}, batch size 4, AdamW), selecting the best checkpoint by validation accuracy.

Results. CLM outperforms MLM at every context length and position (+10.7pp overall: 62.2% vs 51.6%). MLM accuracy degrades monotonically with document

length (57% at 512 tokens to 43% at 8,192), while CLM degrades more slowly and remains above MLM throughout. The CLM advantage is largest at mid-document positions, where the inserted fact is farthest from both sequence boundaries. This is consistent with the downstream pattern: the French tasks with the largest CLM gains (FrACCO, CANTEMIST) use sequence lengths of 4,096 tokens during fine-tuning.

9.5 Analysis: Why Does the CLM Detour Work?

9.5.1 CKA Methodology

We measure representational similarity with linear Centered Kernel Alignment (Kornblith et al., 2019). Given two representation matrices $X \in \mathbb{R}^{n \times p}$ and $Y \in \mathbb{R}^{n \times q}$ (one per model, n samples, mean-pooled over tokens), we center each column, compute Gram matrices $K = XX^\top$ and $L = YY^\top$, and define:

$$\mathrm{CKA}(X, Y) = \frac{\mathrm{tr}(KL)}{\sqrt{\mathrm{tr}(K^2)\,\mathrm{tr}(L^2)}}$$

We report divergence $d = 1 - \mathrm{CKA}$, so higher values indicate greater representational difference. CKA equals 1 when representations are identical up to invertible linear transformation, 0 when there is no linear relationship. All computations use float64 arithmetic. For French we use 500 held-out texts from DiaMED and FrACCO; for English, PubMed abstracts. Results are averaged over 3 seeds.

9.5.2 The CLM Imprint Persists in Low Layers

Comparing layer-by-layer CKA between CLM-detour and MLM-only models, the CLM phase modifies low-layer representations in a way that subsequent MLM training does not undo, even when the decay budget matches the CLM phase. CKA divergence reaches approximately 56.5% after 1.5B tokens of MLM decay and remains stable, with only 0.6pp variation over 8B additional tokens.

To isolate where CLM differs from MLM *specifically* rather than just from training noise, we normalize each layer's CLM-MLM divergence by a seed-noise control (two MLM models trained with different random seeds on identical data):

$$r_\ell = \frac{1 - \mathrm{CKA}(\mathrm{CLM}_\ell,\ \mathrm{MLM}_\ell^{s_1})}{1 - \mathrm{CKA}(\mathrm{MLM}_\ell^{s_2},\ \mathrm{MLM}_\ell^{s_1})}$$

A ratio of 1 means no CLM-specific effect. Low layers (0–7) show ratios of 5–44×: CLM changes them far more than random seed variation does. Mid and deep layers are near 1×: both objectives modify them similarly, so CKA cannot distinguish CLM from MLM there.

Table 9.5: **Freeze intervention results** (French Base, 8 tasks, 9 seeds). Freezing low layers during CLM eliminates the benefit ($p = 0.36$ vs MLM baseline). Freezing mid layers preserves it. Freezing low layers during decay has negligible effect.

Condition	Avg F_1	Δ vs MLM
References		
MLM baseline	57.9	—
CLM detour	**60.8**	+2.9
Freeze interventions		
Freeze L0–7 (CLM phase)	58.4	+0.5
Freeze L8–14 (CLM phase)	60.3	+2.4
Freeze L0–7 (decay phase)	60.3	+2.4

Table 9.6: **Layer transplantation** on DiaMED (3 seeds, linear probing). No transplant recovers more than 35% of the 7.1pp gap.

Transplant	F_1 (%)	Δ	Recovery
References			
MLM baseline	32.8	—	0%
CLM detour	39.9	+7.1	100%
Layer group transplants			
Low (0–7)	31.9	−0.9	−12%
Mid (8–14)	35.3	+2.5	+35%
Late (15–21)	32.4	−0.4	−6%

9.5.3 Causal Evidence: Freeze Interventions

Do these low-layer changes actually drive the downstream improvement? We answer with the freeze interventions described in §9.3.2, summarized in Table 9.5.

Freezing low layers (0–7) during CLM drops F_1 from 60.8% to 58.4%. We cannot reject the null hypothesis that the resulting model matches the MLM baseline ($p = 0.36$, paired bootstrap across 8 tasks × 9 seeds). Freezing mid layers (8–14) during CLM preserves the full benefit (60.3%). Together, these two experiments establish a double dissociation: low layers are necessary and mid layers are not. Freezing low layers during decay has negligible effect (−0.5pp, 60.3%), confirming that MLM decay preserves the CLM imprint in low layers.

9.5.4 Non-Localizability

The freeze experiments show that low layers are necessary *during training*. We now ask whether the resulting benefit can be localized *after training*, by copying full parameter blocks from the CLM model into the MLM model.

Table 9.7: **Component-level transplants** on DiaMED (3 seeds, linear probing). Neither attention nor MLP modules alone recover the full gap.

Transplant	F_1 (%)	Δ	Recovery
All attention	33.9	+1.1	+15%
All MLP	35.3	+2.5	+35%

Table 9.8: **CKA divergence scales with model size and is asymmetric.** The encoder-decoder comparison uses different model families (see text).

Model	Params	Div.	CI
Encoders (MLM→CLM→MLM)			
ModernCamemBERT Base	150M	56.5%	±0.7%
ModernCamemBERT Large	350M	67.2%	±0.3%
Decoder (CLM→MLM→CLM)			
Gemma-3	270M	26.3%	±1.0%

No transplant recovers more than 35% of the gap (Table 9.6). Low and late layer transplants hurt performance. Inserting CLM low layers into an MLM model creates a mismatch with the mid and deep layers that co-adapted under MLM, yielding representations worse than either pure model.

Component-level transplants (Table 9.7) confirm the same pattern. Replacing all attention modules from CLM into MLM recovers only 15% of the gap; replacing all MLP blocks recovers 35%. No single component type captures the full benefit.

Unlike freeze interventions, which act during training, transplants replace already-trained weights and cannot reconstruct the cross-layer co-adaptation that developed during CLM. The freeze experiments show low layers are necessary during training; the transplant experiments show the resulting benefit cannot be isolated post hoc.

9.5.5 Scaling and Architectural Asymmetry

The CLM imprint increases with model capacity (Table 9.8): Large (350M) reaches 67.2% ± 0.3% divergence versus 56.5% ± 0.7% for Base (150M). Testing the reverse on Gemma-3 (270M decoder), MLM-then-CLM retains only 26.3% ± 1.0% divergence versus 67.2% for an encoder of similar scale. One explanation is the density of the decay signal: MLM decay at 15% masking provides weak erasure, while CLM decay at 100% provides strong erasure. This comparison is exploratory: it is confounded by model family and language, and a controlled comparison using encoder and decoder variants of the same family would be needed to confirm the asymmetry.

9.6 Limitations

Our freeze interventions use French Base models only. The CKA patterns that motivate them (low-layer divergence well above seed noise) are consistent across French Base, French Large, and English Base, but replication of the freeze interventions on English and Large models would strengthen the causal claim. The encoder-decoder asymmetry comparison is confounded by model family. The position trade-off at 2048+ tokens means that the approach may not be suitable for tasks requiring very long context, such as full discharge summary coding.

Conclusion

Training an encoder like a decoder, then converting it back, produces stronger representations than training with MLM alone. The CLM detour reshapes low transformer layers in ways that subsequent MLM decay does not undo: a lasting imprint that scales with model capacity and produces consistent downstream gains across 8 French and 11 English biomedical tasks.

The practical implication is straightforward: for biomedical encoder pretraining, the CLM detour is a free lunch. Same data, same compute, better results. MLM decay of roughly 10% of the detour length suffices, and the effect is strongest when the domain gap between pretraining and target data is large.

Parts 1 and 2 of this thesis have focused on data and pretraining: how to collect biomedical text, how to filter it, and how to train encoder models on it. But a pretrained model is only useful if it can be deployed on real tasks. In the next part, we turn to adaptation: how to extract clinical information when labeled data is scarce, labels number in the thousands, and the only available text is public.

10 Knowledge-Enriched Pretraining

10.1 Introduction

Medical coding is the task of assigning standardized codes from medical ontologies to clinical documents. It is central to hospital billing, epidemiological surveillance, and clinical research. In France, over 30 million hospital stays per year require accurate coding using the CIM-10 classification (the French adaptation of ICD-10), CCAM procedure codes, and ATC medication codes. This task requires understanding not only medical vocabulary but also the relational structure of medical ontologies: hierarchical links between codes, causal associations, differential diagnoses, and exclusion rules. These relationships are explicitly encoded in ontology graphs but are absent from clinical text corpora.

Pretrained biomedical language models such as BioBERT (Lee et al., 2020b), PubMedBERT (Gu et al., 2022b), and CamemBERT-bio (Touchent et al., 2023) learn medical vocabulary from large text corpora. However, they do not capture the relational structure of medical ontologies, since this structure is not expressed in running text. Knowledge graph approaches such as RDF2Vec (Ristoski and Paulheim, 2016) and Snomed2Vec (Agarwal et al., 2019) encode ontology structure through random walks, but produce static, non-contextualized embeddings that cannot benefit from transformer pretraining. Knowledge-enhanced transformers such as DRAGON (Yasunaga et al., 2022) and KEPLER (Wang et al., 2021) integrate knowledge graph structure into language model pretraining, but they rely on existing text-graph alignment rather than generating new training data from ontology structure alone.

We propose OntoBook, a pretraining method that converts medical ontology structure into training signal for encoder language models. Our approach proceeds in three stages. First, we generate random walks through ontology graphs to produce sequences that capture hierarchical, causal, and differential relationships between medical codes. Second, we reformulate these walks into fluent medical prose using a large language model, creating synthetic textbooks grounded in ontology structure. Third, we train an encoder with two objectives on the same textbook data: masked language modeling and relation prediction between code pairs. We refer to this co-training on shared data as *alignment*, and our experiments show that it is a necessary condition for the method to work.

We evaluate OntoBook on three French medical coding benchmarks and present ablation studies on the contribution of each component. We release 1.3 million LLM-

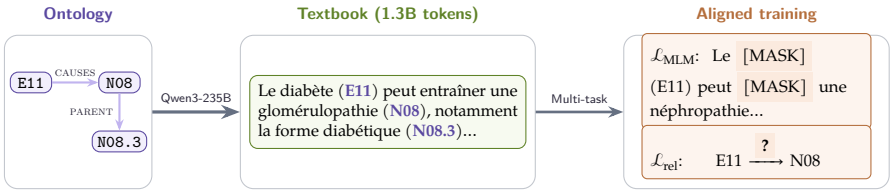

Figure 10.1: **OntoBook pipeline.** An ontology subgraph is converted into a random walk, then reformulated into textbook prose by Qwen3-235B. The encoder trains on this text with two aligned objectives: masked language modeling and relation prediction between code pairs.

reformulated medical textbooks across three French ontologies (CIM-10, CCAM, ATC) and all pretrained model checkpoints.

10.2 Related Work

This section reviews three research directions: biomedical language model pretraining, knowledge-enhanced language models, and automatic medical coding.

10.2.1 Biomedical Pretrained Language Models

Domain-specific pretraining has proven essential for biomedical NLP. BioBERT (Lee et al., 2020b) further pretrained BERT on PubMed abstracts and PMC full-text articles, improving named entity recognition and relation extraction. PubMedBERT (Gu et al., 2022b) demonstrated that pretraining from scratch on in-domain text outperforms mixed-domain initialization. ClinicalBERT (Huang et al., 2019) targeted clinical notes from MIMIC-III. For French, CamemBERT-bio (Touchent et al., 2023) adapted CamemBERT (Martin et al., 2020b) to biomedical text, while DrBERT (Labrak et al., 2023) was trained on French biomedical and clinical text from the NACHOS corpus. These models learn from flat text corpora and do not capture the hierarchical structure of medical ontologies.

10.2.2 Knowledge-Enhanced Language Models

Random walks on knowledge graphs generate sequences capturing relational structure. RDF2Vec (Ristoski and Paulheim, 2016) adapted the walk-then-embed paradigm to RDF graphs. In the biomedical domain, Snomed2Vec (Agarwal et al., 2019) applied random walk and Poincaré embeddings to SNOMED-CT for clinical prediction tasks. OWL2Vec* (Chen et al., 2021) combined random walk sequences with lexical and logical features from OWL ontologies. These methods produce static embeddings (Word2Vec-style) and cannot be used for transformer pretraining.

Several approaches inject knowledge graph structure into transformers. ERNIE (Zhang et al., 2019) aligns entity embeddings with text representations. K-BERT (Liu et al., 2020) injects knowledge triples into the input sequence using soft-position indices and a visibility mask. For multi-task pretraining, KEPLER (Wang et al., 2021) combines MLM with TransE-style knowledge embedding on Wikidata. DRAGON (Yasunaga et al., 2022) jointly trains MLM and knowledge graph link prediction, with a biomedical variant on UMLS achieving +3% on MedQA. CODER (Yuan et al., 2022) uses UMLS relation triplets for contrastive pretraining, and SAPBERT (Liu et al., 2021) uses contrastive learning on UMLS synonyms for biomedical entity linking. Closest to the present work, BioOntoBERT (Shashikumar et al., 2023) generates sentences from biomedical ontologies using Onto2Sen templates and pretrains BERT with MLM on the resulting corpus. However, the template-based generation produces formulaic sentences (e.g., "X is a type of Y") rather than fluent prose, and the method uses only MLM without multi-task relation prediction.

Recent work has explored synthetic data generation from knowledge sources. The Phi series (Gunasekar et al., 2023) demonstrated that LLM-generated textbook-quality data enables strong small-model performance. EntiGraph (Yang et al., 2025b) generates entity-centric synthetic data from knowledge sources but targets decoder language models rather than encoder pretraining.

10.2.3 Automatic Medical Coding

Medical coding assigns ICD, procedure, or medication codes to clinical text. CAML (Mullenbach et al., 2018) introduced label-wise attention for explainable predictions. PLM-ICD (Huang et al., 2022) showed that fine-tuning pretrained encoders with label attention directly improves coding performance. Kirchler et al. (2026) recently demonstrated that LLM embeddings improve transferability of EHR-based predictions across countries and coding systems, highlighting the importance of encoding medical knowledge structure for cross-system generalization. These methods improve the classification head or the input representations but rely on generic encoders pretrained on flat text corpora.

Table 10.1 summarizes how OntoBook combines components not jointly present in prior work.

10.3 Method

OntoBook converts medical ontology structure into pretraining signal through three stages: (1) generating random walks through ontology graphs, (2) reformulating these walks into fluent textbook prose using a large language model, and (3) training an encoder with multi-task learning combining masked language modeling and

Method	Walks	LLM	MLM+Rel
Prior work			
Snomed2Vec	✓		
BioOntoBERT	✓		
DRAGON			✓
KEPLER			✓
Phi-1		✓	
Ours			
OntoBook	✓	✓	✓

Table 10.1: **Comparison with related approaches.** OntoBook uniquely combines all three components for biomedical encoder pretraining.

relation prediction. We first describe the ontologies used, then detail each stage. Figure 10.1 illustrates the full pipeline.

10.3.1 Medical Ontologies

We use three French medical ontologies published by the Agence du Numérique en Santé (ANS) through its terminology server (SMT) in RDF/OWL format. CIM-10 FR PMSI is the French adaptation of the WHO's ICD-10, enriched by the Agence Technique de l'Information sur l'Hospitalisation (ATIH) for the French hospital information system (Programme de Médicalisation des Systèmes d'Information, PMSI). It contains 19,161 diagnostic codes organized in a 5-level hierarchy (chapters, blocks, categories, subcategories), with textual annotations including 23,282 synonyms, 8,171 inclusion notes, and 381 definitions. CIM-10 is the only ontology with semantic edges beyond the hierarchy: 1,317 causal edges (e.g., E11 type 2 diabetes *causes* N08.3 diabetic glomerulopathy), 5,739 exclusion edges, and 341 manifestation edges, yielding a mean node degree of 2.77 compared to 2.00 for the purely hierarchical ontologies.

CCAM (Classification Commune des Actes Médicaux) is the French procedure classification maintained by the Caisse Nationale d'Assurance Maladie (CNAM), containing 38,191 procedure codes in a 4-level hierarchy with 8,991 synonyms and 1,188 disjointness constraints between mutually exclusive procedures. ATC (Anatomical Therapeutic Chemical) is the WHO medication classification with 6,950 substance codes in a strict 5-level hierarchy from anatomical group to chemical substance, with no semantic edges, no synonyms, and no definitions beyond labels. This contrast in relational richness directly shapes the walk generation strategy: CIM-10 walks can follow causal and differential diagnosis paths, whereas CCAM and ATC walks are limited to hierarchical and sibling traversals. Table 10.2 summarizes the structural properties of the three ontologies.

	CIM-10	CCAM	ATC
Graph structure			
Codes	19,161	38,191	6,950
Hierarchy depth	5	4	5
Hierarchical edges	19,139	38,191	6,950
Causal edges	1,317	—	—
Manifestation edges	341	—	—
Exclusion edges	5,739	—	—
Disjoint edges	—	1,188	—
Mean degree	2.77	2.06	2.00
Textual attributes			
Synonyms	23,282	8,991	0
Inclusion notes	8,171	—	—
Definitions	381	151	0

Table 10.2: **Structural properties** of the three French medical ontologies used for walk generation. CIM-10 is the only ontology with semantic edges beyond the hierarchy.

10.3.2 Ontology Walk Generation

We generate random walks through ontology graphs to capture relational structure in a sequential format suitable for language model pretraining. We sample walks starting from each code in the ontology. At each step, the next node is selected via weighted sampling where edge type weights depend on the walk type. This biased sampling is necessary because semantic edges are rare (Table 10.2), and a uniform walk would rarely traverse them. Each walk records the traversed codes along with their labels and the relation types connecting them. Walk length is sampled uniformly between 8 and 20 steps. We track visited nodes to avoid cycles, with a fallback that allows revisiting nodes when all neighbors have been explored.

For CIM-10, we define five walk types, each emphasizing different aspects of medical knowledge: *Etiologie* (causal chains), *Diagnostic Différentiel* (codes to distinguish clinically), *Codage Double* (mandatory code pairs linking etiology to manifestation), *Syndrome* (multi-system manifestations), and *Cross-Chapter* (connections between different ICD chapters, e.g., diabetes E11 to its renal complication N08). We generate 402,328 walks for CIM-10 (average 4,830 characters), 762,958 for CCAM, and 138,980 for ATC, totaling 1,304,266 walks.

10.3.3 Textbook Generation

Raw walks contain structured markers (e.g., "`>> À distinguer de:`") that are useful for parsing but not natural for language model pretraining. We use a large language model to reformulate walks into fluent medical prose resembling textbook paragraphs.

Structured Walk

```
[F02.0] Démence de la maladie de Pick
>> Cette pathologie peut provoquer:
[G31.0] Atrophie cérébrale circonscrite
>> Parmi les causes possibles:
[F02.02] ...avec symptômes hallucinatoires
```

↓ *Qwen3-235B*

Medical Textbook

Le code **F02.0**, désignant la démence de la maladie de Pick, s'inscrit dans un réseau de connexions inter-systèmes entre les troubles mentaux et les affections neurologiques. Ce code entretient une relation étroite avec **G31.0** (atrophie cérébrale circonscrite), établissant un lien étiologique réciproque...

Figure 10.2: **Walk-to-textbook transformation.** The structured walk is reformulated into fluent medical prose while preserving all codes and relations.

We prompt Qwen3-235B-A22B-Instruct (Yang et al., 2025a) with FP8 quantization to transform each walk into a coherent paragraph. The prompt instructs the model to: (1) preserve all medical codes and their relationships exactly as stated, (2) use natural medical language without lists or bullet points, (3) not add any information beyond what appears in the walk. This ensures the textbook content remains grounded in the ontology structure. We apply two filters: we discard reformulations shorter than 50 characters or that fail to mention the source codes. The reformulation runs on 4 NVIDIA H100 GPUs using vLLM (Kwon et al., 2023) with guided decoding to ensure valid JSON output, taking approximately 20 hours for all 1.3 million walks. The resulting textbooks contain approximately 1.3 billion tokens across all three ontologies. Figure 10.2 shows an example transformation.

Figure 10.3 shows the system prompt used for CIM-10 walks (translated from French for readability; prompts for CCAM and ATC follow the same structure with ontology-specific terminology).

10.3.4 OntoBook Training

We train a ModernCamemBERT-base encoder (Antoun et al., 2025), initialized from a French checkpoint trained on 1 trillion tokens.

The model optimizes two objectives jointly. The first, $\mathcal{L}_{\text{MLM}}$, follows standard masked language modeling (Devlin et al., 2019b): we mask 15% of tokens from textbook paragraphs and train to predict them. The second, $\mathcal{L}_{\text{rel}}$, classifies the relationship between pairs of medical codes. Our main model uses CIM-10; we explore transfer from CCAM and ATC in Section 10.4.5. We extract 282,907 code pairs from the CIM-10 ontology across 6 relation types: Parent, Child, Sibling,

Reformulation Prompt (CIM-10, translated)

You are an expert medical writer specializing in high-quality educational material.

Reformulate the provided CIM-10 walk into fluent, textbook-quality medical prose.

Strict rules:
(1) **No invention**: never add information not in the source.
(2) **Total fidelity**: preserve all definitions, codes, clinical notes, exclusions.
(3) **Completeness**: do not omit any source information.
(4) **Ignore metadata**: never mention walk types or structural headers.

Style: Continuous narrative prose, no bullet points or markdown. Integrate CIM-10 codes naturally. Preserve exact medical terminology.

Goal: Transform the walk into text suitable for a medical nosology textbook chapter while remaining 100% faithful to the original content.

Figure 10.3: System prompt for LLM-based walk reformulation. Temperature is set to 0 for deterministic, faithful output.

Differential Diagnosis, Causes, and Caused_By. For each pair, we retrieve the corresponding textbook descriptions generated in Section 10.3.3. The input format is "[CLS] text_A [SEP] text_B [SEP]", where text_A and text_B are the textbook descriptions of the two codes, and a separate linear classification head on the [CLS] token predicts the relation type, trained with cross-entropy loss over the 6 classes.

The combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda \mathcal{L}_{\text{rel}} \tag{10.1}$$

where $\lambda = 1.0$. We train for 4 epochs with learning rate 2×10^{-5}, batch size 128, and AdamW optimizer with weight decay 0.01 and 1,000 warmup steps. Training takes approximately 4 hours per epoch on one NVIDIA H100 GPU. The key design choice is that both objectives operate on the *same textbook data*. We call this property *alignment*, and we show in Section 10.4.3 that it is essential for performance.

10.4 Experiments

10.4.1 Experimental Setup

Evaluation Benchmarks. We evaluate on the standard French medical coding evaluation suite:

- **FRACCO** (Pignat et al., 2025): a native French oncology corpus with 1,301 clinical cases annotated with ICD-O-3.1 codes, framed as multi-label classification over the top 100 codes.

- **Cantemist-FR** (Zaghir et al., 2024): the French adaptation of the Spanish tumor coding task (Miranda-Escalada et al., 2020), with 2,051 clinical documents.

Model	FRACCO	Cant.	Dist.	Avg.
External baselines				
CamemBERT-bio	20.2$_{\pm 0.2}$	12.1$_{\pm 0.4}$	9.0$_{\pm 0.2}$	13.8
DrBERT	36.3$_{\pm 0.7}$	37.7$_{\pm 1.0}$	22.5$_{\pm 0.7}$	32.1
ModernCamemBERT	56.4$_{\pm 1.0}$	63.5$_{\pm 1.3}$	23.4$_{\pm 1.7}$	47.7
Our models				
CodeInfill+MLM	56.5$_{\pm 0.2}$	66.4$_{\pm 1.9}$	20.0$_{\pm 0.9}$	47.6
MLM-only	55.8$_{\pm 0.4}$	66.0$_{\pm 1.1}$	24.2$_{\pm 5.8}$	48.7
OntoBook	**58.3**$_{\pm 0.3}$	**67.1**$_{\pm 1.1}$	**32.2**$_{\pm 1.1}$	**52.5**

Table 10.3: **Main results** on French medical coding (micro-F_1 %). External baselines averaged over 9 seeds; our models over 3 seeds. Best results in **bold**.

- **Distemist-FR** (Zaghir et al., 2024): the French adaptation of the disease mention coding task (Miranda-Escalada et al., 2022).

Cantemist-FR and Distemist-FR are cross-lingual projections from Spanish. FRACCO is the only native French benchmark. All results are micro-F1 scores averaged over 3 random seeds.

Baselines. We compare against French biomedical language models: CamemBERT-bio (Touchent et al., 2023), DrBERT (Labrak et al., 2023), and ModernCamemBERT (Antoun et al., 2025) (which serves as our initialization checkpoint). We also report results for three ablations of our method: MLM-only (trained with $\mathcal{L}_{\text{MLM}}$ only on our textbooks), Rel-only (trained with $\mathcal{L}_{\text{rel}}$ only), and CodeInfill+MLM, which replaces relation prediction with a code infilling objective that masks medical code tokens (e.g., "E11", "N08.3") in textbook text and trains the model to predict them. We do not compare against knowledge-enhanced models such as DRAGON (Yasunaga et al., 2022) or SAPBERT (Liu et al., 2021) as they are English encoders that cannot be applied to French text.

Fine-tuning. For downstream evaluation, each pretrained encoder receives a clinical document as input and predicts a set of medical codes (multi-label classification over the top 100 codes per benchmark). We fine-tune with a linear classification head for 10 epochs, learning rate 2×10^{-5}, and batch size 16. External baselines (CamemBERT-bio, DrBERT, ModernCamemBERT) are averaged over 9 seeds; our models over 3 seeds due to the large number of configurations evaluated.

10.4.2 Main Results

Table 10.3 compares OntoBook against French biomedical baselines.

Configuration	FRACCO	Cant.	Dist.	Avg.	Δ
Loss ablations					
OntoBook (aligned)	**58.33**	**67.06**	**32.24**	**52.54**	—
− $\mathcal{L}_{\text{rel}}$ (MLM-only)	55.81	66.01	24.23	48.68	−3.86
− $\mathcal{L}_{\text{MLM}}$ (Rel-only)	48.24	51.15	20.18	39.86	−12.68
− Alignment (misaligned)	33.39	20.11	12.63	22.04	−30.50
Data ablation					
MLM-only (raw walks)	55.51	66.00	22.76	48.09	−4.45

Table 10.4: **Ablation study** (micro-F_1 %). Misalignment degrades all benchmarks, with the largest drop on Cantemist (−46.95). All variants use the same base model and training budget.

OntoBook outperforms all baselines on all three benchmarks. The improvement over MLM-only pretraining is significant on FRACCO (+2.5 micro-F1, $p < 0.001$) and Distemist (+8.0 micro-F1, $p < 0.001$), with consistent but not individually significant gains on Cantemist ($p = 0.11$). The gain is largest on Distemist, suggesting that ontology-aware pretraining particularly helps with fine-grained disease coding. OntoBook also reduces variance on Distemist from $\pm$5.8 (MLM-only) to $\pm$1.1.

10.4.3 Ablation Study

Table 10.4 ablates the key components of OntoBook.

Alignment is the most critical factor. Training $\mathcal{L}_{\text{MLM}}$ and $\mathcal{L}_{\text{rel}}$ on different data causes catastrophic degradation across all benchmarks, with Cantemist dropping from 67.06 to 20.11 (−46.95). Relation prediction alone also fails (−12.68 average F1), confirming that $\mathcal{L}_{\text{rel}}$ cannot serve as a standalone pretraining objective. The last row trains MLM-only on raw structured walks, removing both reformulation and relation prediction. The raw walk model (48.09) is close to the textbook MLM-only model (48.68), indicating that reformulation alone contributes little to MLM pretraining (+0.59 F1). However, the gap to full OntoBook is 4.45 points, reflecting the combined effect of adding both reformulation and relation prediction. This suggests that reformulation primarily benefits the relation prediction objective: on raw walks with structural markers (e.g., "`>> À distinguer de:`"), the classifier can exploit formatting shortcuts rather than learning medical semantics, whereas fluent textbook prose forces the model to learn from content rather than format.

10.4.4 Training Dynamics and Hyperparameter Sensitivity

Figure 10.4 shows training dynamics and hyperparameter sensitivity. Epoch 1 (51.98) and epoch 2 (52.54) both outperform MLM-only pretraining (dashed line), but epoch 3 (49.43) degrades below it, possibly due to overfitting on the relation prediction

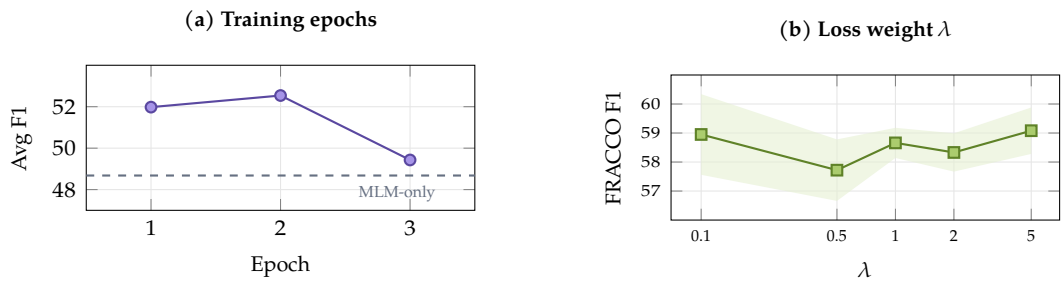

Figure 10.4: **Training dynamics.** (a) Average F_1 peaks at epoch 2 then degrades. (b) Performance is stable across λ values (shaded: ± 1 std).

Ontology source	FRACCO	Cant.	Dist.	Avg.
Main configuration				
CIM-10 (OntoBook)	58.33	67.06	32.24	52.54
Other ontology sources				
All (CIM-10+CCAM+ATC)	59.31	**67.64**	31.44	52.80
CCAM only	58.59	66.54	34.26	**53.13**
ATC only	**59.66**	63.03	**36.05**	52.91

Table 10.5: **Effect of ontology source** on downstream performance (micro-F_1 %). Each row trains a separate model using walks and relations from only that ontology. All models use the same training procedure and base checkpoint.

task. All results in this paper use the epoch 2 checkpoint. The loss weight λ is robust across $\{0.1, 0.5, 1.0, 2.0, 5.0\}$: all values achieve 57.7–59.1% F1 on FRACCO. We use $\lambda = 1.0$ as it achieves the lowest variance ($\pm 0.52\%$). This robustness suggests that the exact balance between objectives matters less than ensuring both are present and aligned.

10.4.5 Multi-Ontology Transfer

The main OntoBook model uses CIM-10 walks (Table 10.3). To test whether other ontologies also provide useful pretraining signal, we train separate models on walks from each ontology individually and on all three combined.

All three individual ontologies improve over the MLM-only baseline, indicating that the pipeline generalizes across different medical knowledge structures. ATC (medications) yields the best Distemist score (+3.81 over CIM-10), despite Distemist being a disease coding task, suggesting that drug-disease relationships transfer well to disease understanding. CCAM (procedures) achieves the best average micro-F1 (53.13). Combining all three ontologies does not outperform the best individual ontologies. We hypothesize that this is because the relation prediction head must accommodate heterogeneous edge types across ontologies.

Model	Chapter	Depth	Dist. ρ
Baselines			
Base	97.5	99.3	0.195
MLM-only	98.9	99.4	0.287
CodeInfill+MLM	**99.0**	**99.8**	**0.397**
Rel-only	98.0	98.2	0.203
Ours			
OntoBook	98.7	99.7	0.073

Table 10.6: **Probing analysis** on CIM-10 code embeddings (% accuracy for Chapter and Depth, Spearman ρ for Distance). CodeInfill+MLM achieves the best geometric organization but OntoBook achieves the best downstream performance (Table 10.3).

10.4.6 Probing Analysis

To understand how OntoBook organizes code representations, we evaluate with three probing tasks on CIM-10 code embeddings (mean-pooled over code descriptions): chapter classification (26 classes), hierarchy depth prediction (4 levels), and distance correlation (Spearman ρ between cosine distances and ontology tree distances). We compare against a code infilling variant (CodeInfill+MLM) that masks and predicts code tokens during pretraining.

CodeInfill+MLM achieves the best probing metrics ($\rho = 0.397$) but the worst downstream performance among our models (47.6 average micro-F1, Table 10.3), while OntoBook has lower distance correlation ($\rho = 0.073$) despite achieving the best downstream performance (52.5 micro-F1). This inverse relationship suggests that for encoder pretraining, representational flexibility is more valuable than strict geometric preservation of ontology structure.

10.5 Discussion

The 30.50-point gap between aligned and misaligned training highlights the central role of objective alignment. When $\mathcal{L}_{\text{MLM}}$ trains on textbooks while $\mathcal{L}_{\text{rel}}$ trains on unrelated data, the model receives conflicting gradient signals: the MLM objective pushes representations toward textbook language patterns, while the relation objective pushes them toward a different data distribution. Aligned training avoids this conflict by ensuring both objectives reinforce the same semantic structure. Relation prediction guides attention toward ontologically meaningful patterns in the textbook text, while MLM ensures the model builds coherent representations of that text. This explains why relation prediction alone fails (−12.68 F1): it is a discriminative task that can exploit surface-level shortcuts, as demonstrated by the small gap between raw-walk and textbook MLM-only models (+0.59 F1). Combined with aligned MLM,

however, it provides a complementary signal that improves downstream coding performance.

The cross-ontology transfer results suggest that our approach generalizes beyond CIM-10: training on CCAM or ATC alone also improves over the baseline despite targeting different medical domains. However, combining all ontologies does not outperform individual ones, possibly because the relation prediction head must accommodate heterogeneous edge types.

The probing analysis reveals a tension between geometric organization and downstream performance. CodeInfill+MLM achieves the best probing metrics ($\rho = 0.397$) but the worst downstream scores among our models (47.6 micro-F1), while OntoBook achieves the opposite ($\rho = 0.073$, 52.5 micro-F1). This suggests that for downstream coding tasks, flexible representations adapt better than rigid geometric structure that must be partially unlearned during fine-tuning.

LLM reformulation plays a dual role. For MLM alone, the effect is modest (+0.59 F1), but reformulation is essential for relation prediction: on raw walks, structural markers provide formatting shortcuts that the classifier can exploit without learning medical semantics, whereas fluent prose forces the model to learn from content rather than format. The full gap between OntoBook and MLM-only on raw walks is 4.45 points.

Limitations. All experiments use French medical coding. Extending to English ontologies such as ICD-11 and SNOMED-CT is left for future work. The reformulation step requires significant compute for LLM inference, and sensitivity to the choice of LLM has not been evaluated. Cantemist-FR and Distemist-FR are cross-lingual projections from Spanish (Zaghir et al., 2024), which may introduce translation artifacts; FRACCO is the only native French benchmark. All experiments use ModernCamemBERT; generalization to other architectures remains untested.

Future Work. Future work will extend OntoBook to English medical coding using UMLS and SNOMED-CT ontologies. A promising direction is cross-ontology walks that traverse edges between different ontologies (e.g., from a CIM-10 diagnosis to its ATC treatment to the CCAM procedure), generating richer walks that capture inter-ontology relationships in a single sequence.

10.6 Conclusion

We presented OntoBook, a method that converts medical ontology structure into pretraining signal for encoder language models through random walks, LLM-based textbook reformulation, and multi-task learning combining masked language modeling and relation prediction. Our key finding is that alignment between objectives

is essential: both tasks must train on the same data, as misalignment causes a 30-point degradation. On French medical coding benchmarks, OntoBook improves over MLM-only pretraining by +2.5 micro-F1 on FRACCO and +8.0 on Distemist. We release 1.3 million LLM-reformulated medical textbooks across three French ontologies (CIM-10, CCAM, ATC) and pretrained model checkpoints.

Part IV

Adapting to Clinical Tasks

11 Limits of Direct Fine-Tuning

12 Architectures for Low-Resource Extraction

13 Synthetic Data for Task Adaptation

14 Conclusion

Acronyms

PCA	Principal component analysis
SNF	Smith normal form
TDA	Topological data analysis

Glossary

LaTeX	A document preparation system
$\mathbb{R}$	The set of real numbers

Bibliography

Khushbu Agarwal, Tome Eftimov, Raghavendra Addanki, Sutanay Choudhury, Suzanne Tamang, and Robert Rallo. 2019. Snomed2vec: Random walk and poincaré embeddings of a clinical knowledge base for healthcare analytics. In *Proceedings of the KDD Workshop on Applied Data Science for Healthcare* (*DSHealth*).

Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. 2019. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of KDD*, pages 2623–2631.

Wissam Antoun, Benoît Sagot, and Djamé Seddah. 2025. ModernBERT or DeBERTaV3? Examining architecture and data influence on transformer encoder models performance. In *Proceedings of the International Joint Conference on Natural Language Processing and the Asia-Pacific Chapter of the Association for Computational Linguistics* (*IJCNLP-AACL 2025*).

Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. SciBERT: A pretrained language model for scientific text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3615–3620. Association for Computational Linguistics.

Aman Berhe, Guillaume Draznieks, Vincent Martenot, Valentin Masdeu, Lucas Davy, and Jean-Daniel Zucker. 2023. ALIBERT: A pretrained language model for French biomedical text. HAL hal-03911564.

Rémi Cardon, Natalia Grabar, Cyril Grouin, and Thierry Hamon. 2020. Présentation de la campagne d'évaluation DEFT 2020. In *Actes de TALN 2020, Atelier DÉfi Fouille de Textes*, pages 1–13. ATALA et AFCP.

Jiaoyan Chen, Pan Hu, Ernesto Jimenez-Ruiz, Ole Magnus Holter, Denvar Antonyrajah, and Ian Horrocks. 2021. OWL2Vec*: Embedding of OWL ontologies. *Machine Learning*, 110(7):1813–1845.

Zeming Chen, Alejandro Hernández Cano, Angelika Romanou, et al. 2023. Meditron-70B: Scaling medical pretraining for large language models. *arXiv preprint arXiv:2311.16079*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451. Association for Computational Linguistics.

Jenny Copara, Julien Knafou, Nona Naderi, Claudia Moro, Patrick Ruch, and Douglas Teodoro. 2020. Contextualized French Language Models for Biomedical Named Entity Recognition. In *Actes de TALN 2020, Atelier DÉfi Fouille de Textes*, pages 36–48. ATALA.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019a. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4171–4186. Association for Computational Linguistics.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019b. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1* (*Long and Short Papers*), pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Basile Dura, Charline Jean, Xavier Tannier, Alice Calliger, Romain Bey, Antoine Neuraz, and Rémi Flicoteaux. 2022. Learning structures of the French clinical language. Technical report, arXiv. ArXiv:2207.12940.

Ettin et al. 2025. Bridging the encoder-decoder gap with continued pretraining. *arXiv preprint*.

Hippolyte Gisserot-Boukhlef et al. 2025. Should we still pretrain encoders with masked language modeling? *arXiv preprint arXiv:2507.00994*.

Nathan Godey, Wissam Antoun, Rian Touchent, Rachel Bawden, Éric de la Clergerie, Benoît Sagot, and Djamé Seddah. 2026. GAPeron: A large-scale french language model. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. To appear.

Natalia Grabar and Rémi Cardon. 2018. CLEAR – Simple Corpus for Medical French. In *Proceedings of the 1st Workshop on Automatic Text Adaptation*, pages 3–9. Association for Computational Linguistics.

Aaron Grattafiori, Abhimanyu Dubey, et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Cyril Grouin, Natalia Grabar, Vincent Claveau, and Thierry Hamon. 2019. Clinical case reports for NLP. In *Proceedings of the 18th BioNLP Workshop and Shared Task*, pages 273–282. Association for Computational Linguistics.

Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2022a. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23.

Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2022b. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23.

Suriya Gunasekar, Yi Zhang, Jyoti Anber, Sébastien Bubeck, Ronen Eldan, Neel Ganapathy, Yin Tat Lee, Yuanzhi Li, et al. 2023. Textbooks are all you need. *arXiv preprint arXiv:2306.11644*.

Chao-Wei Huang, Shang-Chi Tsai, and Yun-Nung Chen. 2022. PLM-ICD: Automatic ICD coding with pretrained language models. In *Proceedings of the 4th Clinical Natural Language Processing Workshop*, pages 10–20. ACL.

Kexin Huang, Jaan Altosaar, and Rajesh Ranganath. 2019. Clinicalbert: Modeling clinical notes and predicting hospital readmission. *arXiv preprint arXiv:1904.05342*.

Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. 2021. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. In *Applied Sciences*.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. 2019. PubMedQA: A dataset for biomedical research question answering. *arXiv preprint arXiv:1909.06146*.

Diederik P. Kingma and Jimmy Ba. 2015. Adam: A method for stochastic optimization. ICLR 2015. arXiv:1412.6980.

Matthias Kirchler, Marcelo Ferro, Valentina Lorenzini, et al. 2026. Large language models improve transferability of electronic health record-based predictions across countries and coding systems. *npj Digital Medicine*.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of neural network representations revisited. In *ICML*.

Taku Kudo and John Richardson. 2018. SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 66–71. Association for Computational Linguistics.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th ACM Symposium on Operating Systems Principles*, pages 611–626. ACM.

Yanis Labrak, Adrien Bazoge, Richard Dufour, Béatrice Daille, Pierre-Antoine Gourraud, Emmanuel Morin, and Mickael Rouvier. 2022. FrenchMedMCQA: A French multiple-choice question answering dataset for medical domain. In *Proceedings of the LREC 2022 Workshop on Multilingual Biomedical Text Processing*.

Yanis Labrak, Adrien Bazoge, Richard Dufour, Mickael Rouvier, Emmanuel Morin, Béatrice Daille, and Pierre-Antoine Gourraud. 2023. DrBERT: A robust pre-trained model in French for biomedical and clinical domains. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 16207–16221. Association for Computational Linguistics.

Yanis Labrak, Adrien Bazoge, Emmanuel Morin, Pierre-Antoine Gourraud, Mickael Rouvier, and Richard Dufour. 2024. BioMistral: A collection of open-source pretrained large language models for medical domains. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics.

Tiphaine Le Clercq de Lannoy, Romaric Besançon, Olivier Ferret, Julien Tourille, Frédérique Brin-Henry, and Bianca Vieru. 2022. Stratégies d'adaptation pour la reconnaissance d'entités médicales en français. In *Actes de la 29e Conférence sur le Traitement Automatique des Langues Naturelles*, pages 215–225. ATALA.

Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020a. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240.

Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020b. BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240.

Jeffrey Li, Alex Fang, Georgios Smyrnis, et al. 2024. DataComp-LM: In search of the next generation of training sets for language models. *arXiv preprint arXiv:2406.11794*.

D. A. Lindberg, B. L. Humphreys, and A. T. McCray. 1993. The Unified Medical Language System. *Methods of Information in Medicine*, 32(4):281–291.

Fangyu Liu, Ehsan Shareghi, Zaiqiao Meng, Marco Basaldella, and Nigel Collier. 2021. Self-alignment pretraining for biomedical entity representations. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4228–4238. ACL.

Weijie Liu, Peng Zhou, Zhe Zhao, Zhiruo Wang, Qi Ju, Haotang Deng, and Ping Wang. 2020. K-BERT: Enabling language representation with knowledge graph. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 2901–2908.

Bernardo Magnini, Begoña Altuna, Alberto Lavelli, Manuela Speranza, and Roberto Zanoli. 2020. The E3C project: Collection and annotation of a multilingual corpus of clinical cases. *Proceedings of the Seventh Italian Conference on Computational Linguistics CLiC-it 2020*.

Louis Martin, Benjamin Muller, Pedro Javier Ortiz Suárez, Yoann Dupont, Laurent Romary, Éric de la Clergerie, Djamé Seddah, and Benoît Sagot. 2020a. CamemBERT: a tasty French language model. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7203–7219, Online. Association for Computational Linguistics.

Louis Martin, Benjamin Muller, Pedro Javier Ortiz Suárez, Yoann Dupont, Laurent Romary, Éric de la Clergerie, Djamé Seddah, and Benoît Sagot. 2020b. CamemBERT: A tasty French language model. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7203–7219. ACL.

Antonio Miranda-Escalada, Eulàlia Farré-Maduell, and Martin Krallinger. 2020. Named entity recognition, concept normalization and clinical coding: Overview of the Cantemist track for cancer text mining in Spanish, corpus, guidelines, methods and results. In *Proceedings of the Iberian Languages Evaluation Forum (IberLEF 2020), CEUR Workshop Proceedings*, volume 2664, pages 303–323.

Antonio Miranda-Escalada, Luis Gascó, Salvador Lima-López, Eulàlia Farré-Maduell, Darryl Estrada, Anastasios Nentidis, Anastasia Krithara, Georgios Katsimpras, Georgios Paliouras, and Martin Krallinger. 2022. Overview of DisTEMIST at BioASQ: Automatic detection and normalization of diseases from clinical texts: Results, methods, evaluation and multilingual resources. In *Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2022), CEUR Workshop Proceedings*, volume 3180, pages 179–203.

James Mullenbach, Sarah Wiegreffe, Jon Duke, Jimeng Sun, and Jacob Eisenstein. 2018. Explainable prediction of medical codes from clinical text. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1101–1111. ACL.

Hiroki Nakayama. 2018. seqeval: A Python framework for sequence labeling evaluation. Https://github.com/chakki-works/seqeval.

Aurélie Névéol, K. Bretonnel Cohen, Cyril Grouin, Thierry Hamon, Thomas Lavergne, Liadh Kelly, Lorraine Goeuriot, Grégoire Rey, Aude Robert, Xavier Tannier, and Pierre Zweigenbaum. 2016. Clinical information extraction at the CLEF eHealth evaluation lab 2016. *CEUR Workshop Proceedings*, 1609:28–42.

Aurélie Névéol, Cyril Grouin, Jeremy Leixa, Sophie Rosset, and Pierre Zweigenbaum. 2014. The QUAERO French medical corpus: A resource for medical entity recognition and normalization. In *Proc. of BioTextMining Workshop*, pages 24–30.

Pedro Javier Ortiz Suárez, Benoît Sagot, and Laurent Romary. 2019. Asynchronous pipelines for processing huge corpora on medium to low resource infrastructures. In *Proceedings of the Workshop on Challenges in the Management of Large Corpora* (*CMLC-7*), pages 9–16. Leibniz-Institut für Deutsche Sprache.

Ankit Pal, Logesh Kumar Umapathi, and Malaikannan Sankarasubbu. 2022. MedMCQA: A large-scale multi-subject multi-choice dataset for medical domain question answering. In *Proceedings of the Conference on Health, Inference, and Learning*.

Guilherme Penedo, Hynek Kydlíček, Loubna Cappelli, Hailey Saez de Ocáriz Borde, et al. 2024. FineWeb: decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*.

Johann Pignat, Milena Vucetic, Christophe Gaudet-Blavignac, Jamil Zaghir, Amandine Stettler, Fanny Amrein, Jonatan Bonjour, Jean-Philippe Goldman, Olivier Michielin, Christian Lovis, and Mina Bjelogrlic. 2025. FRACCO: A gold-standard annotated corpus of oncological entities with ICD-O-3.1 normalisation.

Preethi Raghavan, James L. Chen, Eric Fosler-Lussier, and Albert M. Lai. 2014. How essential are unstructured clinical narratives and information fusion to clinical trial recruitment? *AMIA Summits on Translational Science Proceedings*, 2014:218–223.

David Rein, Betty Li Hou, Asa Cooper Stickland, et al. 2023. GPQA: A graduate-level google-proof q&a benchmark. *arXiv preprint arXiv:2311.12022*.

Petar Ristoski and Heiko Paulheim. 2016. RDF2Vec: RDF graph embeddings for data mining. In *International Semantic Web Conference*, pages 498–514. Springer.

Supreeth P Shashikumar, Fatemeh Amrollahi, and Shamim Nemati. 2023. Leveraging biomedical ontologies to boost pre-training for language models. In *Proceedings of the International Conference on Biomedical Ontologies* (*ICBO*), volume 3603 of *CEUR Workshop Proceedings*.

Thomas Sounack et al. 2025. BioClinical ModernBERT: State-of-the-art long context biomedical and clinical encoder. *arXiv preprint arXiv:2506.10896*.

AntLM Team. 2024. AntLM: Bridging causal and masked language models. In *CoNLL*.

OLMo Team, Dirk Groeneveld, et al. 2025. OLMo 2: Furious. *arXiv preprint arXiv:2501.00656*.

Rian Touchent, Laurent Romary, and Éric De La Clergerie. 2023. CamemBERT-bio: a tasty French language model better for your health. In *Actes de la 30ème Conférence sur le Traitement Automatique des Langues Naturelles* (*TALN*), pages 323–334.

Erik M. van Mulligen, Zubair Afzal, Saber A. Akhondi, Dang Vo, and Jan A. Kors. 2016. Erasmus MC at CLEF eHealth 2016: Concept recognition and coding in French texts. In *Working Notes of CLEF 2016*.

Xiaozhi Wang, Tianyu Gao, Zhaocheng Zhu, Zhengyan Zhang, Zhiyuan Liu, Juanzi Li, and Jian Tang. 2021. KEPLER: A unified model for knowledge embedding and pre-trained language representation. *Transactions of the Association for Computational Linguistics*, 9:176–194.

Benjamin Warner, Benjamin Clavié, et al. 2024. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Wang, Bowen Zheng, Bowen Yu, et al. 2025a. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Zitong Yang, Neil Band, Shuangping Li, Emmanuel Candès, and Tatsunori Hashimoto. 2025b. Synthetic continued pretraining. In *Proceedings of the International Conference on Learning Representations* (*ICLR*).

Michihiro Yasunaga, Antoine Bosselut, Hongyu Ren, Xikun Zhang, Christopher D Manning, Percy Liang, and Jure Leskovec. 2022. Deep bidirectional language-knowledge graph pretraining. In *Advances in Neural Information Processing Systems*, volume 35, pages 37309–37323.

Zheng Yuan, Zhengyun Zhao, Haixia Sun, Jiao Li, Fei Wang, and Sheng Yu. 2022. CODER: Knowledge-infused cross-lingual medical term embedding for term normalization. *Journal of Biomedical Informatics*, 126:103983.

Jamil Zaghir, Mina Bjelogrlic, Jean-Philippe Goldman, Soukaïna Aananou, Christophe Gaudet-Blavignac, and Christian Lovis. 2024. FRASIMED: A clinical French annotated resource produced through crosslingual BERT-based annotation projection. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation* (*LREC-COLING 2024*), pages 7450–7460, Torino, Italia. ELRA and ICCL.

Zhengyan Zhang, Xu Han, Zhiyuan Liu, Xin Jiang, Maosong Sun, and Qun Liu. 2019. ERNIE: Enhanced language representation with informative entities. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 1441–1451. ACL.

Appendix